ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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PHYSICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Magnetic Effect of Current

English Medium

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

A. Biot-Savart law + Ampere's law

1. A moving charge produces
(A) an electric field only (B) a magnetic field only
(C) both electric and magnetic fields (D) neither an electric nor magnetic field

MG0313

2. One Wb/m^2 is equal to

(A) 10^4 gauss (B) 10^2 gauss (C) 10^{-2} gauss (D) 10^{-4} gauss

MG0314

3. A piece of wire carrying a current of 6.00 A is bent in the form of a circular arc of radius 10.0 cm, and it subtends an angle of 120° at the centre. Find the magnetic field B due to this piece of wire at the centre.
(A) zero (B) 1.26×10^{-5} T (C) 5×10^{-5} T (D) 7.2×10^{-5} T

MG0040

4. The magnetic field due to a current carrying circular loop of radius 3 cm at a point on the axis at a distance of 4 cm from the centre is 54 μT. What will be its value at the centre of the loop ?
(A) 250 μT (B) 150 μT (C) 125 μT (D) 75 μT

MG0043

5. A circular current -carrying coil has radius R. The distance from the centre of the coil, on the axis, where B will be 1/8 of its value at the centre of the coil is

(A) $R/\sqrt{3}$ (B) $\sqrt{3}R$ (C) $2\sqrt{3}R$ (D) $\frac{2R}{\sqrt{3}}$

MG0315

6. Two similar coils of radius R and number of turn N are lying concentrically with their planes at right angles to each other. The current flowing in them are I and 2I respectively. The resultant magnetic induction at the centre will be–

(A) $\frac{\mu_0 NI}{R}$ (B) $\frac{\mu_0 NI}{2R}$ (C) $\frac{\sqrt{3}\,\mu_0 NI}{2R}$ (D) $\frac{\sqrt{5}\,\mu_0 NI}{2R}$

MG0316

7. A charge q coulomb is circulating in an orbit of radius r meters making n revolutions per second. The magnetic field produced at the centre of the circle is

(A) $\frac{2\pi q}{nr}\times 10^{-7}$ (B) $\frac{2\pi q}{r}\times 10^{-7}$ (C) $\frac{2\pi nq}{r}\times 10^{-7}$ (D) $\frac{2\pi rn}{q}\times 10^{-7}$

MG0317

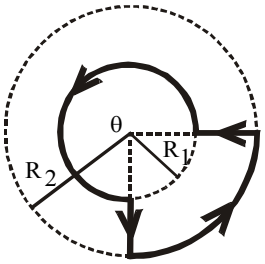
8. An electron of charge e coulomb is going around in an orbit of radius R metres in a hydrogen atom with velocity v m/s. The magnetic flux density associated with it at the centre of the atom is

(A) $\frac{\mu_0 ev}{4\pi R^2}$ (B) $\frac{\mu_0 ev}{2R}$ (C) $\mu_0 evR$ (D) $\frac{\mu_0 ev}{R^2}$

MG0318

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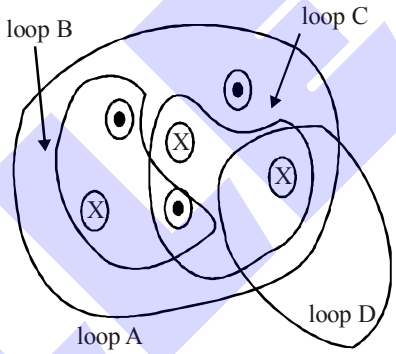
9. What is the magnetic field at the common centre of the wire circuit (continuous line) as shown in the figure below ?



(A) $\frac{\mu_0 I}{8}\left[\frac{1}{R_1}+\frac{1}{R_2}\right]$ (B) $\frac{\mu_0 I}{8}\left[\frac{3}{R_1}+\frac{1}{R_2}\right]$ (C) $\frac{\mu_0 I}{8}\left[\frac{1}{R_1}+\frac{3}{R_2}\right]$ (D) $\frac{\mu_0 I}{8}\left[\frac{3}{R_1}-\frac{1}{R_2}\right]$

MG0319

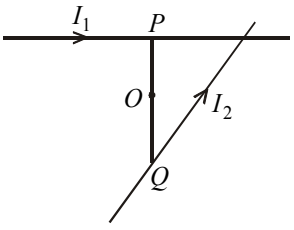
10. Consider six wires coming into or out of the page, all with the same current. Rank the line integral of the magnetic field (from most positive to most negative) taken counterclockwise around each loop shown.



(A) B > C > D > A (B) B > C = D > A (C) B > A > C = D (D) C > B = D > A

MG0047

11. Two long straight wires, carrying currents I_1 and I_2, are placed perpendicular to each other. The line of shortest distance between them. PQ, has length r. The magnitude of the magnetic field at the centre O of PQ is



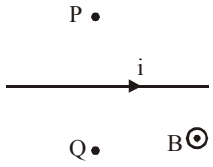
(A) $\frac{2\sqrt{2}\mu_0}{\pi r}(I_1+I_2)$ (B) $\frac{2\sqrt{2}\mu_0}{\pi r}(I_1-I_2)$

(C) $\frac{\mu_0}{\pi r}\left(I_1^2+I_2^2\right)^{1/2}$ (D) $\frac{\mu_0}{2\pi r}\left(I_1^2+I_2^2\right)^{1/2}$

MG0320

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12. A long, straight wire carrying a current of 1.0 A is placed horizontally in a uniform magnetic field $B = 1.0 \times 10^{-5}$ T pointing vertically upward (figure). The magnitude of the resultant magnetic field at the points P and Q, both situated at a distance of 2.0 cm from the wire in the same horizontal plane are respectively



(A) zero, 20 μT (B) 20 μT, zero (C) zero, zero (D) 20 μT, 20 μT

MG0036

13. Two long straight wires, each carrying a current I, are separated by a distance r. If the currents are in opposite directions, then the strength of the magnetic field at any point midway between the two wires is

(A) $\frac{\mu_0 I}{\pi r}$ (B) $\frac{2\mu_0 I}{\pi r}$ (C) $\frac{\mu_0 I}{2\pi r}$ (D) zero

MG0321

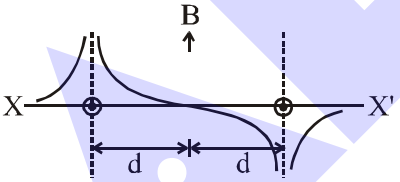
14. Two parallel long straight conductors are placed at right angles to a metre scale at the 2 cm and 6 cm marks. They carry currents I and 3I respectively in the same direction. The resultant magnetic field due to them is zero.

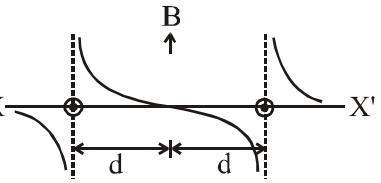
(A) at the 3 cm mark (B) at the 4 cm mark
(C) at the 5 cm mark (D) nowhere

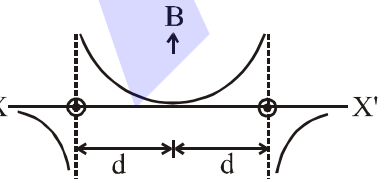
MG0322

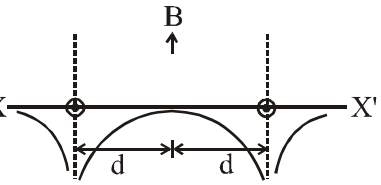
15. Two long parallel wires are at a distance 2d apart. They carry steady equal currents flowing out of the plane of the paper as shown. The variation of the magnetic field B along the line XX' is given by:-

[AIEEE - 2010]

(A)


(B)


(C)


(D)


MG0166

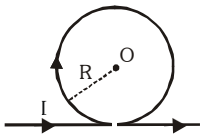
16. A current of 1A is flowing in the sides of an equilateral triangle of side $4.5\times10^{-2} m$. The magnetic field at the centroid of the triangle is

(A) $2\times10^{-5} T$ (B) $4\times10^{-5} T$ (C) $8\times10^{-5} T$ (D) $1.2\times10^{-4} T$

MG0323

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17. An infinitely long straight conductor is bent into the shape as shown in figure. It carries a current I ampere and the radius of the circular loop is r meter. Then the magnetic induction at the centre of the circular part is :



(A) Zero (B) ∞ (C) $\frac{\mu_0}{4\pi}\frac{2I}{r}(\pi+1)$ (D) $\frac{\mu_0}{4\pi}\frac{2I}{r}(\pi-1)$

MG0324

18. A long straight wire of radius a carries a steady current i. The current is uniformly distributed across its cross-section. The ratio of the magnetic field at $\frac{a}{2}$ and 2a is-

(A) $\frac{1}{4}$ (B) 4 (C) 1 (D) $\frac{1}{2}$

MG0053

19. Two co-axial wires one solid and one hollow, each carry the same current I in opposite directions. The radius of inner wire is r_1. The inner radius of the hollow wire is r_2 and the outer radius is r_3. What is the magnetic induction at a distance r from the axis, such that $r_2 < r < r_3$?

(A) $\frac{\mu_0 I}{2\pi r}[r_3^2 - r^2]$ (B) $\frac{\mu_0 I}{2\pi r}[r_3^2 - r_2^2]$ (C) $\frac{\mu_0 I}{2\pi r}\left[\frac{r_3^2 - r^2}{r_3^2 - r_2^2}\right]$ (D) $\frac{\mu_0 I}{2\pi r}\left[\frac{1}{r_3^2 - r_2^2}\right]$

MG0325

20. In the above question, what is magnetic induction at a point where $r_1 < r < r_2$?

(A) $\frac{\mu_0 I}{2\pi r}$ (B) $\frac{\mu_0 I}{2\pi r}[r_2^2 - r_1^2]$ (C) $\frac{\mu_0 I}{2\pi r}[r_2^2 - r_1^2]$ (D) $\frac{\mu_0 I}{2\pi r}\left[\frac{1}{r_2^2 - r_1^2}\right]$

MG0326

21. In the above question, what is the magnetic field at a point where $r < r_1$?

(A) $\frac{\mu_0}{2\pi}\left[\frac{Ir_1}{r^2}\right]$ (B) $\frac{\mu_0}{2\pi}\left[\frac{Ir}{r_1^2}\right]$ (C) $\frac{\mu_0}{2\pi}[I]$ (D) $\frac{\mu_0}{2\pi}\left[\frac{Ir^2}{r_1^2}\right]$

MG0327

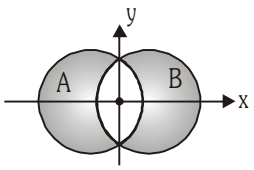
22. In the above question, what is the magnetic field at a point where $r > r_3$?

(A) $\frac{\mu_0}{4\pi}\left[\frac{I}{r}\right]$ (B) $\frac{\mu_0}{2\pi}\left[\frac{I}{r}\right]$ (C) $\frac{\mu_0}{2\pi}\left[\frac{2I}{r}\right]$ (D) zero

MG0328

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23. Two cylindrical straight and very long non magnetic conductors A and B, insulated from each other, carry a current I in the positive and the negative z–direction respectively. The direction of magnetic field at origin is



(A) $-\hat{i}$ (B) $+\hat{i}$

(C) $\hat{j}$ (D) $-\hat{j}$

MG0329

24. A closely wound solenoid 80 cm long has 5 layers of windings of 400 turns each. The diameter of the solenoid is 1.8 cm. If the current carried is 8.0 A. estimate the magnitude of B inside the solenoid near its centre. ***(NCERT)***

(A) zero (B) $8\pi \times 10^{-3}$ T (C) $15\pi \times 10^{-3}$ T (D) $\pi \times 10^{-3}$ T

MG0051

25. The magnetic field inside a current carrying toroidal solenoid is B. It its radius is doubled and the current through it is also doubled, the magnetic field inside the solenoid will be

(A) B/2 (B) B (C) 4 A (D) 4 B

MG0330

B. Magnetic force on a charged particle

26. An electric charge + q moves with velocity $\vec{V} = 3\hat{i} + 4\hat{j} + \hat{k}$, in an electromagnetic field given by

$\vec{E} = 3\hat{i} + \hat{j} + 2\hat{k}$, $\vec{B} = \hat{i} + \hat{j} - 3\hat{k}$

The y-component of the force experienced by + q is :- **[AIEEE - 2010]**

(A) 2 q (B) 11 q (C) 5 q (D) 3q

MG0168

27. If magnetic force on a charge is given by $q(\vec{v} \times \vec{B})$ and a charged particle is projected in a magnetic field $(2\hat{i} + 2\hat{j} + 2\hat{k})$ tesla. The acceleration of the particle at an instant is $(x\hat{i} + 2\hat{j} - 6\hat{k})$ m/s². Value of x is

(A) 4 (B) 2 (C) 3 (D) 1

MG0331

28. An electron is injected into a uniform magnetic field with components of velocity parallel to and normal to the field direction. The path of the electron is a

(A) helix (B) circle (C) parabola (D) straight line

MG0332

29. A beam of electrons is accelerated through a potential difference V. It is then passed normally through a uniform magnetic field where it moves in a circle of radius r. It would have moved in a circle of radius $2r$, if it were initially accelerated through a potential difference

(A) $\sqrt{2}V$ (B) $2V$ (C) $2\sqrt{2}V$ (D) $4V$

MG0333

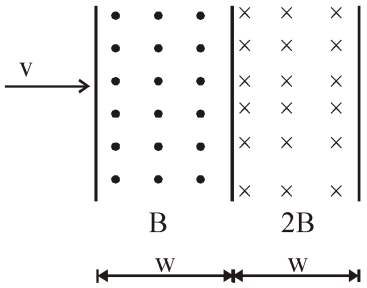
30. A proton and an α-particle, accelerated through the same potential difference, enter a region of uniform magnetic field normally. If the radius of the proton orbit is 10 cm, then that of the α-orbit is

(A) 10 cm (B) $10\sqrt{2}$ *cm* (C) 20 cm (D) $5\sqrt{2}$ *cm*

MG0334

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31. The magnetic field shown in the figure consist of the two magnetic fields. If v is the velocity just required for a charge particle of mass m and charge q to pass through the magnetic field. Particle is projected with velocity "v" then how much time does such a charge spend in the magnetic field -



(A) $\frac{\pi m}{2qB}$ (B) $\frac{\pi m}{qB}$

(C) $\frac{\pi m}{4qB}$ (D) $\frac{3\pi m}{2qB}$

MG0335

32. In a cyclotron, a charged particle
(A) undergoes acceleration all the time.
(B) speeds up between the dees because of the magnetic field.
(C) speeds up in a dee.
(D) slows down within a dee and speeds up between dees.

MG0077

33. A tightly-wound, long solenoid carries a current of 2.00 A. An electron is found to execute a uniform circular motion inside the solenoid with a frequency of 1.00×10^8 rev/s. Find the number of turns per metre in the solenoid.
(A) 500 Turns/m (B) 1020 Turns/m (C) 1232 Turns/m (D) 1420 Turns/m

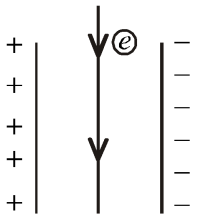
MG0061

34. A charge particle is moving inside both electric field and magnetic field without any acceleration. Electric field is $10^3 \hat{i}\frac{N}{C}$ and magnetic field is $2 \times 10^{-2}\ \hat{j}$ T. The velocity of charge particle is–

(A) $5\times10^{+4}\hat{k}$
(B) $2\times10^{3}\hat{k}$
(C) $4\times10^{4}(-\hat{k})$
(D) May be both $5\times10^{-4}\hat{k}$ and $5\times10^{-4}(-\hat{k})$ depend on nature of charge

MG0336

35. A potential difference of 600 V is applied across the plates of a parallel plate capacitor. The separation between the plates is 3 mm. A magnetic field also exists between the plates. An electron projected parallel to the plates as shown with a speed of $2\times10^{6}\ m/s$ moves undeflected between the plates. The magnitude and direction of the magnetic field is



(A) 0.2 T, into the page (B) 0.2 T out of the page
(C) 0.1 T into the page (D) 0.1 T out of the page

MG0337

36. A proton moving with a constant velocity passes through a region of space without any change in its velocity. If E and B represent the electric and magnetic fields respectively, this region of space cannot be

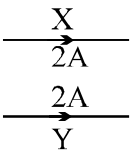
(A) $E=0,\ B=0$ (B) $E=0,\ B\neq0$ (C) $E\neq0,\ B=0$ (D) $E\neq0,\ B\neq0$

MG0338

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C. Magnetic force on a current carrying wire

37. In given figure, X and Y are two long straight parallel conductors each carrying a current of 2 A. The force on each conductor is F newtons. When the current in each is changed to 1 A and reversed in direction, the force on each is now



(A) F/4 and unchanged in direction
(B) F/2 and reversed in direction
(C) F/2 and unchanged in direction
(D) F/4 and reversed in direction

MG0078

38. A conducting circular loop of radius r carries a constant current i. It is placed in a uniform magnetic field B such that B is perpendicular to the plane of the loop. The magnetic force acting on the loop is

(A) $ri\,B$ (B) $2\pi\,ri\,B$ (C) zero (D) $\pi\,ri\,B$

MG0339

39. A current of 3 A is flowing in a linear conductor having a length of 40 cm. The conductor is placed in a magnetic field of strength 500 gauss and makes an angle of 30° with the direction of the field. It experiences a force of magnitude

(A) $3\times10^{4}\,N$ (B) $3\times10^{2}\,N$ (C) $3\times10^{-2}\,N$ (D) $3\times10^{-4}\,N$

MG0340

40. A 0.5 m wire, stretched horizontally, carries a current of 10 A from east to west in a magnetic field of 0.1 T directed vertically downwards. The magnitude and direction of the force on the wire are

(A) 0.5 N towards south
(B) 0.5 N towards north
(C) 0.25 N towards south
(D) 0.25 N towards north

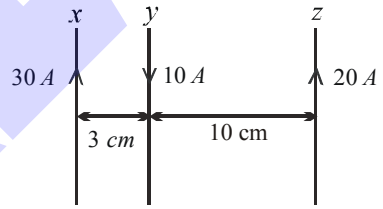
MG0341

41. One ampere is the current which when flowing through each of two parallel wires spaced 1m apart in vacuum and of infinite length will give rise to a force between them equal to

(A) $1\,N/m$ (B) $1\times10^{-2}\,N/m$ (C) $2\times10^{-7}\,N/m$ (D) $4\pi\times10^{-7}\,N/m$

MG0342

42. In the figure, x, y and z are long straight wires. The magnetic force on 25 cm length of y is



(A) $2\times10^{-4}\,N$ towards right
(B) $4\times10^{-4}\,N$ towards right
(C) $2\times10^{-4}\,N$ towards left
(D) $4\times10^{-4}\,N$ towards left

MG0343

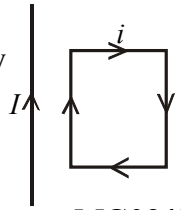
43. A metal rod of mass 10 g and length 10 cm is placed on a smooth 60° incline. A uniform magnetic field B is applied in the region vertically downwards. If a current of 3 A is passed through the rod, then the value of B for which the rod remains stationary on the inclined plane is

(A) $3T$ (B) $\frac{1}{3}T$ (C) $\sqrt{3}\,T$ (D) $\frac{1}{\sqrt{3}}T$

MG0344

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44. A rectangular loop carrying a current i is situated near a long straight wire such that the wire is parallel to one of the sides of the loop and is in the plane of the loop. If a steady current I is established in the wire as shown in the figure, the loop will



(A) rotate about an axis parallel to the wire (B) move away from the wire
(C) move towards are wire (D) remain stationary

MG0345

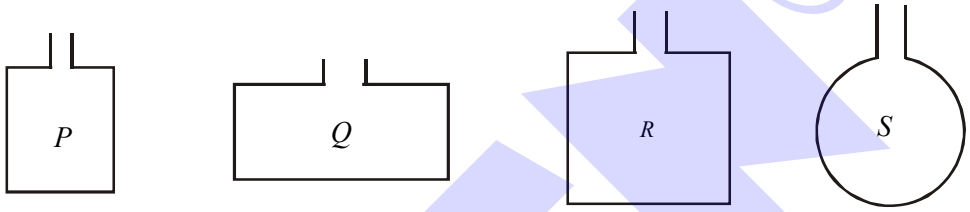
D. Torque on a current carrying loop

45. A magnetic needle lying parallel to a magnetic field requires W unit of work to turn it through 60°. The torque needed to maintain the needle in this position will be–

(A) $\sqrt{3}$ W (B) W (C) ($\sqrt{3}$ /2)W (D) 2W

MG0346

46. Four wires, each of length 2.0 m, are bent into four loops P, Q, R and S and then suspended in a uniform magnetic field. If the same current passes in each, then the torque will be maximum in the loop



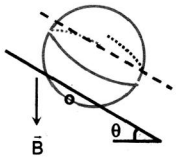
(A) P (B) Q (C) R (D) S

MG0347

47. In the figure shown, a coil of single turn is wound on a sphere of radius r and mass m. The plane of the coil is parallel to the inclined plane and lies in the equatorial plane of the sphere. If the sphere is in rotational equilibrium, the value of B is (current in the coil is i)

(A) $\frac{mg}{\pi ir}$ (B) $\frac{mg\sin\theta}{\pi i}$

(C) $\frac{mgr\sin\theta}{\pi i}$ (D) None of these



MG0348

48. A thin circular disk of radius R is uniformly charged with density $\sigma > 0$ per unit area. The disk rotates about its axis with a uniform angular speed ω. The magnetic moment of the disk is :-

[AIEEE - 2011]

(A) $2\pi R^4\sigma\omega$ (B) $\pi R^4\sigma\omega$ (C) $\frac{\pi R^4}{2}\sigma\omega$ (D) $\frac{\pi R^4}{4}\sigma\omega$

MG0169

E. Magnetic material + Earth magentism

49. The susceptibility of a diamagnetic material

(A) decreases with temperature
(B) does not vary with temperature
(C) first decreases and then increases with temperature
(D) increases with temperature

MG0349

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current-EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Exp\65

50. When a magnet is heated, it
(A) loses its magnetism
(B) gains magnetism
(C) gains magnetism up to a certain temperature and loses magnetism beyond that temperature
(D) neither gains nor loses magnetism

MG0350

51. A material satisfied the relation $\mu_0(H + M) = 0$, where H and M are magnetic intensity and magnetisation respectively, then material is :-
(A) nonmagnetic (B) paramagnetic (C) ferrormagnetic (D) diamagnetic

MG0114

52. **Statement-1 :** The sensitivity of a moving coil galvanometer is increased by placing a suitable magnetic material as a core inside the coil.
and
Statement-2 : Soft iron has a high magnetic permeability and cannot be easily magnetized or demagnetized.
(A) Statement-1 is True, Statement-2 is True ; statement-2 is a correct explanation for statement-1
(B) Statement-1 is True, Statement-2 is True ; statement-2 is NOT a correct explanation for statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True

MG0116

53. The area of the B-H hysteresis loop is an indication of the
(A) permeability of the material (B) susceptibility of the material
(C) retentivity of the material (D) energy dissipated per cycle

MG0351

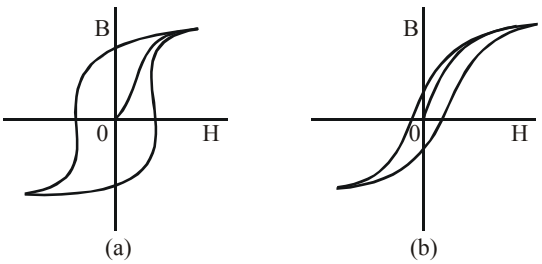
54. Alnico is preferred for making permanent magnets due to its
(A) high retentivity, high coercivity (B) high retentivity, low coercivity
(C) low retentivity, high coercivity (D) low retentivity, low coercivity

MG0352

55. Soft iron is preferred as the core of transformers due to its
(A) high retentivity, high coercivity (B) high retentivity, low coercivity
(C) low retentivity, high coercivity (D) low retentivity, low coercivity

MG0353

56. Two types of steel are characterized by the hysteresis loops shown in figure (a) and (b). The loops are obtained in the processes of magnetization and demagnetization of the steels. Which of the two types is best suited for using as the core of a transformer and which, for using as a permanent magnet ?



(A) Both are best suited for transformer
(B) Both are best suited for permanent magnet
(C) A is best suited for permanent magnet & B is best suited for core of transformer
(D) A is best suited for core of transformer & B is best suited for permanent magnet

MG0113

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57. The coercivity of a bar magnet is 100A/m. It is to be demagnetised by placing it inside a solenoid of length 100 cm and number of turns 50. The current flowing the solenoid will be
(A) 4A (B) 2A (C) 1A (D) zero

MG0354

58. Declination is the angle between
(A) geographic and magnetic meridians
(B) geographic meridian and horizontal component of earth's magnetic field
(C) horizontal and vertical components of earth's magnetic field
(D) horizontal component and total magnetic field of the earth

MG0355

59. A bar magnet is placed in the north-south direction with its north pole towards north. In which direction from the centre of the magnet will the points of zero magnetic field lie ?
(A) North-South (B) East-West
(C) North-East - South-West (D) North-West - South-East

MG0356

60. The line on the earth's surface joining the points where the earth's field is horizontal is called the
(A) magnetic meridian (B) magnetic axis (C) magnetic line (D) magnetic equator

MG0357

61. At Geo-magnetic poles, the angle of dip is :-
(A) 45° (B) 30° (C) zero (D) 90°

MG0358

62. At a place, the angle of dip is 30°. If the horizontal component of earth's magnetic field is H, then the total field intensity will be given by

(A) $H/2$ (B) $2H/\sqrt{3}$ (C) $H\sqrt{2}$ (D) $H\sqrt{3}$

MG0359

63. A magnet is suspended in such a way that it oscillates in the horizontal plane. It makes 20 oscillations per minute at a place where dip angle is 30° and 15 oscillations per minute at a place where dip angle is 60°. Ratio of the total earth's magnetic field at the two places is

(A) $3\sqrt{3}:8$ (B) $16:9\sqrt{3}$ (C) 4 : 9 (D) $2\sqrt{3}:9$

MG0360

64. At two places A and B using vibration magnetometer, a magnet vibrates in a horizontal plane and its respective periodic time are 2 sec and 3 sec and at these places the earths horizontal components are B_A and B_B respectively. Then the ratio between B_A and B_B will be :-
(A) 9 : 4 (B) 3 : 2 (C) 4 : 9 (D) 2 : 3

MG0361

65. Points A and B are situated on the equatorial plane of a small bar magnet at distances x and 3x from its centre. The ratio of the magnetic fields at A and B is approximately
(A) 1 : 9 (B) 9 : 1 (C) 27 : 1 (D) 1 : 27

MG0362

66. Two small magnets, each of magnetic moment $10\,A-m^2$, are placed in end-on position 0.1m apart from their centres. The force acting between them is

(A) $0.6\times10^7\,N$ (B) $0.06\times10^7\,N$ (C) $0.6\,N$ (D) $0.06\,N$

MG0363

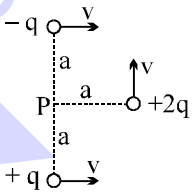
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current-EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Exp\65

EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

A. Biot-Savart law + Ampere's law

1. A charge $+2q$ moves vertically upwards with speed v, a second charge $-q$ moves horizontally to the right with the same speed v, and a third charge $+q$ moves horizontally to the right with the same speed v. The point P is located a perpendicular distance a away from each charge as shown in the figure. The magnetic field at point P is



(A) Into the page with magnitude $\frac{\mu_0}{4\pi}\frac{2qv}{a^2}$

(B) Into the page with magnitude $\frac{\mu_0}{4\pi}\frac{4qv}{a^2}$

(C) Out of the page with magnitude $\frac{\mu_0}{4\pi}\frac{2qv}{a^2}$

(D) Out the page with magnitude $\frac{\mu_0}{4\pi}\frac{4qv}{a^2}$

MG0045

2. Two circular coils 1 and 2 are made from same wire but radius of 1st coil is twice that of 2nd coil. What potential difference should be applied across them so that the magnetic field at their centres is the same ?

(A) 2 (B) 3 (C) 4 (D) 6

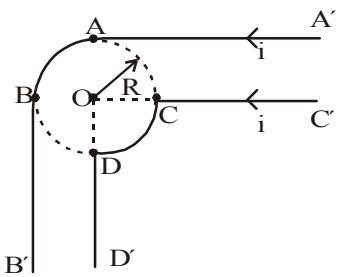
MG0364

3. Considering magnetic field along the axis of a circular loop of radius 1 meter, at what distance from the centre of the loop the magnetic field is $\frac{1}{2\sqrt{2}}$ times of its value at the centre ?

(A) 1 m (B) 3 m (C) 5 m (D) 9 m

MG0046

4. All straight wires are very long. Both AB and CD arc area of the same circle, both subtending right angles at the centre O. Then the magnetic field at O is



(A) $\frac{\mu_0 i}{4\pi R}$ (B) $\frac{\mu_0 i}{4\pi R}\sqrt{2}$

(C) $\frac{\mu_0 i}{2\pi R}$ (D) $\frac{\mu_0 i}{2\pi R}(\pi+1)$

MG0041

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5. A horizontal overhead powerline is at a height of 4m from the ground and carries a current of 100 A from east to west. The magnetic field directly below it one the ground is($\mu_0 = 4\pi \times 10^{-7} TmA^{-1}$)

(A) 2.5×10^{-7} T southward (B) 5×10^{-6} T northward

(C) 5×10^{-6} T southward (D) 2.5×10^{-7} T northward

MG0037

6. Two parallel wires carry equal currents of 10 A along the same direction and are separated by a distance of 2.0 cm. Find the magnetic field at a point which is 2.0 cm away from each of these wires.

(A) 3.4×10^{-4} T in a direction parallel to the plane of the wires and perpendicular to the wires

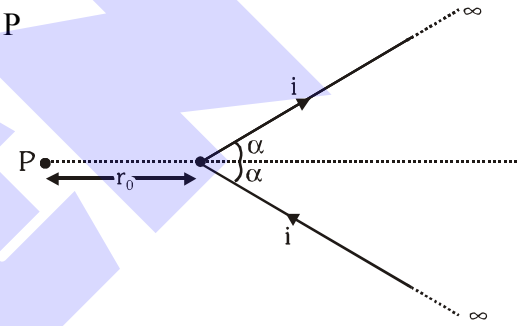
(B) 1.7×10^{-4} T in a direction parallel to the plane of the wires and parallel to the wires

(C) 1.7×10^{-4} T in a direction parallel to the plane of the wires and perpendicular to the wires

(D) 3.4×10^{-4} T in a direction parallel to the plane of the wires and parallel to the wires

MG0038

7. A current carrying wire in the form of 'V' alphabet is kept as shown in the figure. Magnetic field intensity at point P which lies on the angular bisector of V is



(A) $\frac{\mu_0 i}{4\pi r_0}[1-\cos\alpha]$ (B) $\frac{\mu_0 i}{2\pi r_0}[1-\cos\alpha]$

(C) $\frac{\mu_0 i}{4\pi r_0}\frac{[1-\cos\alpha]}{\sin\alpha}$ (D) $\frac{\mu_0 i}{2\pi r_0}\frac{[1-\cos\alpha]}{\sin\alpha}$

MG0365

8. A regular hexagon carries current I. The hexagon can be inscribed inside a circle of radius R. What is the magnetic field at the centre of the circle ?

(A) $\frac{\mu_0}{4}\frac{2I}{R}$ (B) $\frac{\mu_0}{4\pi}\frac{12I}{R}$ (C) $\frac{\mu_0}{4\pi}\frac{I}{R}\sqrt{3}$ (D) $\frac{\mu_0}{4\pi}\frac{12I}{R}\sqrt{3}$

MG0366

9. **Statement-1 :** Ampere law can be used to find magnetic field due to finite length of a straight current carrying wire.

Statement-2 : The magnetic field due to finite length of a straight current carrying wire is symmetric about the wire.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

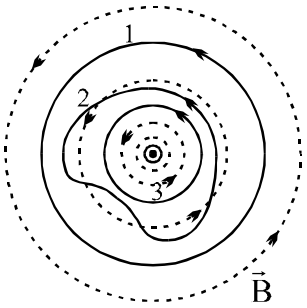
(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

MG0048

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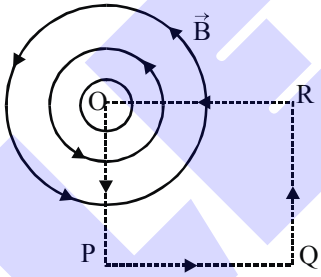
10. Consider the three closed loops drawn using solid line in the magnetic field (magnetic field lines are drawn using dotted line) of an infinite current-carrying wire normal to the plane of paper as shown. Rank the line integral of the magnetic field along each path in order of increasing magnitude



(A) 1 > 2 > 3 (B) 1 = 3 > 2 (C) 1 = 2 = 3 (D) 3 > 2 > 1

MG0055

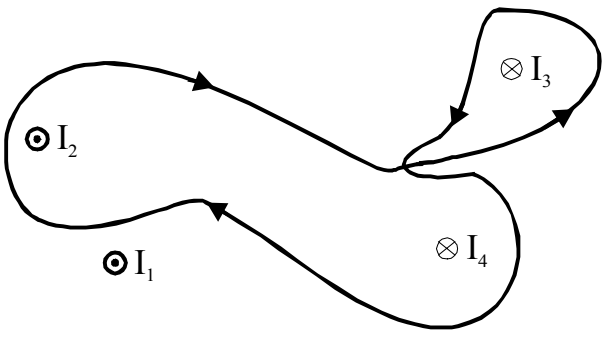
11. A current carrying wire (current = i) perpendicular to the plane of the paper produces a magnetic field, as shown in the figure. A square of side a is drawn with one of its vertices on the centre of the wire. The integral $\int \vec{B}.d\vec{r}$ along $OPQRO$ has the value



(A) $+\mu_0 i$ (B) $\frac{\mu_0 i}{8}$ (C) $\frac{\mu_0 i}{4}$ (D) $\frac{\mu_0 i}{2}$

MG0056

12. Four wires carrying current I_1 = 2A, I_2 = 4A, I_3 = 6A and I_4 = 8A respectively cut the page perpendicularly as shown in figure. The value of $\int \vec{B}.d\vec{\ell}$ for the loop shown would be :-



(A) +2 μ_0-wb/m (B) –2 μ_0-wb/m (C) + 10 μ_0-wb/m (D) – 10 μ_0-wb/m

MG0057

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3.Magnetic Effect of Current+EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Ex.p65

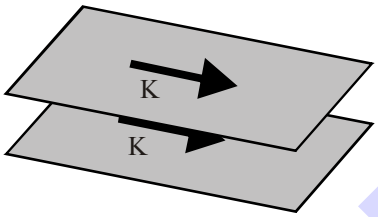
13. A long solenoid has 200 turns per cm and carries a current i. The magnetic field at its centre is 6.28×10^{-2} weber/m^2. Another long solenoid has 100 turns per cm and it carries a current i/3. The value of the magnetic field at its centre is-

(A) 1.05×10^{-2} weber/m^2 (B) 1.05×10^{-5} weber/m^2

(C) 1.05×10^{-3} weber/m^2 (D) 1.05×10^{-4} weber/m^2

MG0052

14. What are the directions of the magnetic field between and outside a pair of two parallel large sheets carrying currents in the same directions, as illustrated in Figure (from the side shown)?



(A) towards us between the plates and away from us above and below the plates.

(B) toward us above the plates and away from us below plates and zero between plates.

(C) towards us above and below the plates and zero between the plates

(D) towards us between the plates and zero above and below the plates.

MG0058

15. Two mutually perpendicular long conducting wire carrying currents I_1 and I_2 lie in one plane. Locus of the point at which the magnetic induction is zero, is a

(A) circle with centre as the point of intersection of the wire.

(B) parabola with vertex as the point of intersection of the wire

(C) straight line passing through the point of intersection of the wire

(D) rectangular hyperbola

MG0089

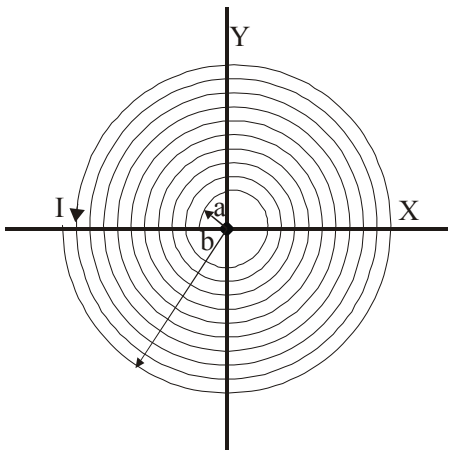
16. A long insulated copper wire is closely wound as a spiral of 'N' turns. The spiral has inner radius 'a' and outer radius 'b'. The spiral lies in the X-Y plane and a steady current '*I*' flows through the wire. The Z-component of the magnetic field at the center of the spiral is **[IIT-JEE 2011]**

(A) $\frac{\mu_0 NI}{2(b-a)}\ell n\left(\frac{b}{a}\right)$

(B) $\frac{\mu_0 NI}{2(b-a)}\ell n\left(\frac{b+a}{b-a}\right)$

(C) $\frac{\mu_0 NI}{2b}\ell n\left(\frac{b}{a}\right)$

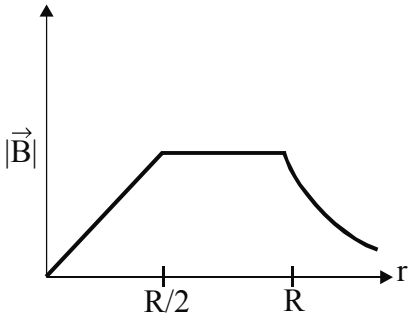
(D) $\frac{\mu_0 NI}{2b}\ell n\left(\frac{b+a}{b-a}\right)$

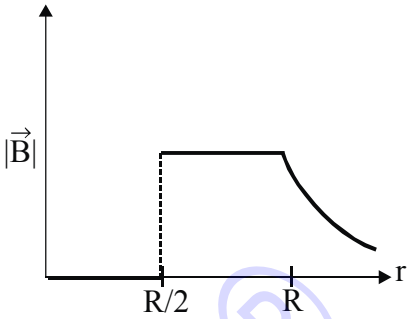


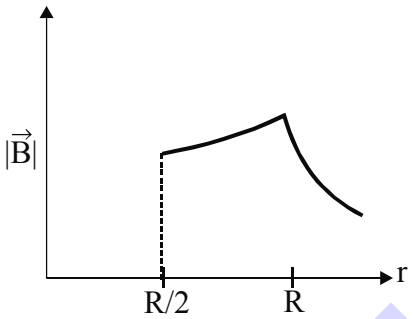
MG0186

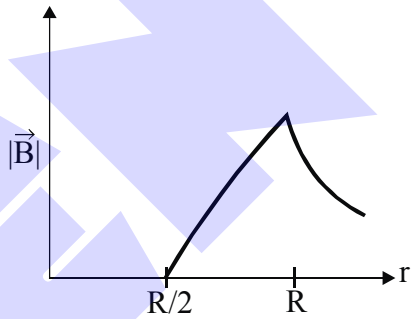
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17. An infinitely long hollow conducting cylinder with inner radius R/2 and outer radius R carries a uniform current density along its length. The magnitude of the magnetic field, $|\vec{B}|$ as a function of the radial distance r from the axis is best represented by **[IIT-JEE 2012]**

(A)


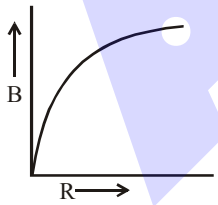
(B)


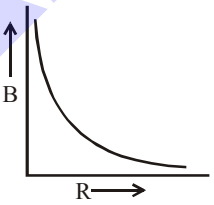
(C)


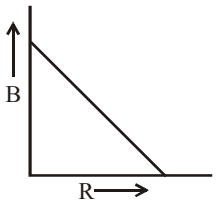
(D)


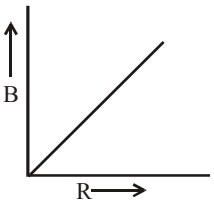
MG0189

18. A charge Q is uniformly distributed over the surface of non-conducting disc of radius R. The disc rotates about an axis perpendicular to its plane and passing through its centre with an angular velocity ω. As a result of this rotation a magnetic field of induction B is obtained at the centre of the disc. If we keep both the amount of charge placed on the disc and its angular velocity to be constant and vary the radius of the disc then the variation of the magnetic induction at the centre of the disc will be represented by the figure :- **[AIEEE-2012]**

(A)


(B)


(C)


(D)


MG0170

B. Magnetic force on a charged particle

19. An electron is moving along positive x-axis. A uniform electric field exists towards negative y-axis. What should be the direction of magnetic field of suitable magnitude so that net force of electron is zero

(A) positive z- axis (B) negative z-axis (C) positive y-axis (D) negative y-axis

MG0063

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3.Magnetic Effect of Current+EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Ex.p65

20. A charged particle moves in a magnetic field $\vec{B} = 10\hat{i}$ with initial velocity $\vec{u} = 5\hat{i} + 4\hat{j}$. The path of the particle will be

(A) straight line (B) circle (C) helical (D) none

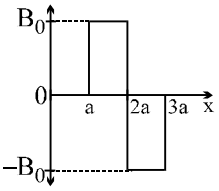
MG0066

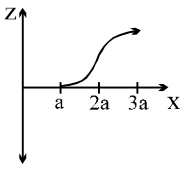
21. A particle of mass m and charge q moves with a constant velocity v along the positive x direction. It enters a region containing a uniform magnetic field B directed along the negative z direction, extending from x = a to x = b. The minimum value of v required so that the particle can just enter the region x > b is :- **JEE 2002 (screening)]**

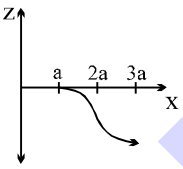
(A) q b B/m (B) q(b – a) B/m (C) q a B/m (D) q(b + a) B/2m

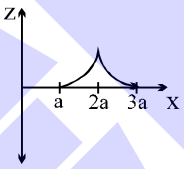
MG0070

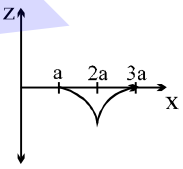
22. A magnetic field $\vec{B} = B_0\hat{j}$ exists in the region a < x < 2a and $\vec{B} = -B_0\hat{j}$, in the region 2a < x < 3a, where B_0 is a positive constant. A positive point charge moving with a velocity $\vec{v} = v_0\hat{i}$, where v_0 is a positive constant, enters the magnetic field at x = a. The trajectory of the charge in this region can be like,



(A) 

(B) 

(C) 

(D) 

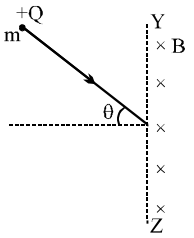
MG0071

23. A particle having charge of 1 C, mass 1 kg and speed 1 m/s enters a uniform magnetic field, having magnetic induction of 1 T, at an angle θ = 30° between velocity vector and magnetic induction. The pitch of its helical path is (in meters)

(A) $\frac{\sqrt{3}\pi}{2}$ (B) $\sqrt{3}\pi$ (C) $\frac{\pi}{2}$ (D) π

MG0073

24. A particle with charge +Q and mass m enters a magnetic field of magnitude B, existing only to the right of the boundary YZ. The direction of the motion of the particle is perpendicular to the direction of B. Let $T = 2\pi\frac{m}{QB}$. The time spent by the particle in the field will be



(A) Tθ (B) 2Tθ (C) $T\left(\frac{\pi+2\theta}{2\pi}\right)$ (D) $T\left(\frac{\pi-2\theta}{2\pi}\right)$

MG0074

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25. In the previous question, if the particle has –Q charge, the time spend by the particle in the field will be :-

(A) $T\theta$ (B) $2T\theta$ (C) $T\left(\frac{\pi+2\theta}{2\pi}\right)$ (D) $T\left(\frac{\pi-2\theta}{2\pi}\right)$

MG0075

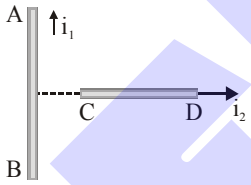
C. Magnetic force on a current carrying wire

26. A wire of mass 100 g carrying a current of 2A towards increasing x is in the form of $y = x^2$ ($-2m \le x \le +2m$). This wire is placed in a magnetic field $B = -0.02\hat{k}$ Tesla & gravity free region. The acceleration of the wire (in m/s^2) is :-

(A) $-1.6\hat{j}$ (B) $-3.2\hat{j}$ (C) $1.6\hat{j}$ (D) $2.4\hat{j}$

MG0081

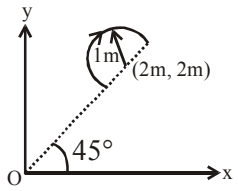
27. An inifinitely long straight conductor AB is fixed and a current is passed through it. Another movable straight wire CD of finite length and carrying current is held perpendicular to it and released. Neglect weight of the wire:



(A) The rod CD will move upwards parallel to itself
(B) The rod CD will move downwarnd parallel to itself
(C) The rod CD will move upward and turn clockwise at the same time
(D) The rod CD will move upward and turn anti-clockwise at the same time.

MG0367

28. A uniform magnetic field $\vec{B} = \left(3\hat{i} + 4\hat{j} + \hat{k}\right)$ Tesla exist in a region of space. A semicircular wire of radius 1 m carrying a current of 1A having its centre at (2,2,0) m is placed on the X-Y plane as shown. The force on the semicircular wire will be



(A) $\frac{1}{\sqrt{2}}\left(\hat{i}-\hat{j}+\hat{k}\right)N$ (B) $\sqrt{2}\left(\hat{i}-\hat{j}+\hat{k}\right)N$

(C) $\frac{1}{\sqrt{2}}\left(\hat{i}+\hat{j}-\hat{k}\right)N$ (D) $\sqrt{2}\left(\hat{i}+\hat{j}-\hat{k}\right)N$

MG0090

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3.Magnetic Effect of Current+EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Ex.p65

29. A very long wire carrying current I is fixed along x-axis. Another parallel finite wire carrying a current in the opposite direction is kept at a distance d above the wire in xy plane. The second wire is free to move parallel to itself. The options available for its small displacements are in

(i) +ve x direction (ii) +ve y direction (iii) +ve z direction

Taking gravity in negative y direction, the nature of equilibrium of second wire is

(A) stable for movement in x direction, unstable for movement in y direction, neutral for movement in z direction

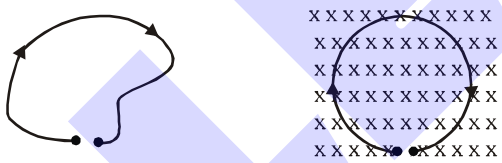
(B) stable for movement in y direction, unstable for movement in z direction, neutral for movement in x direction

(C) stable for movement in z direction, unstable for movement in y direction, neutral for movement in x direction

(D) stable for movement in y direction, unstable for movement in x direction, neutral for movement in z direction

MG0091

30. A thin flexible wire of length L is connected to two adjacent fixed points and carries a current I in the clockwise direction, as shown in the figure. When the system is put in a uniform magnetic field of strength B going into the plane of the paper, the wire takes the shape of a circle. The tension in the wire is : **[2010, 3M]**



(A) IBL (B) $\frac{IBL}{\pi}$ (C) $\frac{IBL}{2\pi}$ (D) $\frac{IBL}{4\pi}$

MG0182

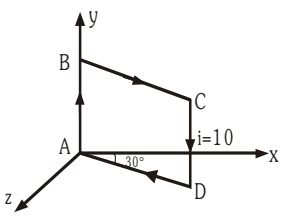
D. Torque on a current carrying loop

31. A circular loop of radius R carries a current I. Another circular loop of radius r(<<R) carries a current i and is placed at the centre of the larger loop. The planes of the two circles are at right angle to each other. Find the torque acting on the smaller loop.

(A) zero (B) $\frac{\mu_0 \pi i I r^2}{4R}$ (C) $\frac{\mu_0 \pi i I r^2}{2R}$ (D) $\frac{\mu_0 \pi i I r^2}{R}$

MG0082

32. Figure shows a square current carrying loop ABCD of side 10 cm and current i = 10 A. The magnetic moment $\vec{M}$ of the loop is



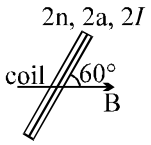
(A) $(0.05)\left(\hat{i}-\sqrt{3}\hat{k}\right)$ A – m^2 (B) $(0.05)\left(\hat{j}+\hat{k}\right)$ A – m^2

(C) $(0.05)\left(\sqrt{3}\hat{i}+\hat{k}\right)$ A – m^2 (D) $\left(\hat{i}+\hat{k}\right)$ A – m^2

MG0368

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current-EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Exp\65

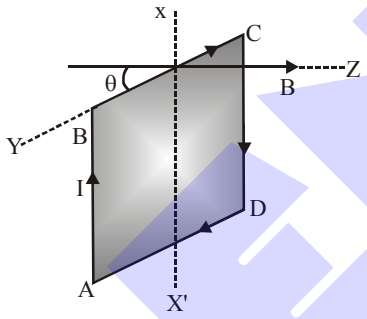
33. A rectangular coil PQ has 2n turns, an area 2a and carries a current 2*I*, (refer figure). The plane of the coil is at 60° to a horizontal uniform magnetic field of flux density B. The torque on the coil due to magnetic force is :-



(A) Bna*I* sin60° (B) 8Bna*I* cos60° (C) 4na*I* Bsin60° (D) none

MG0084

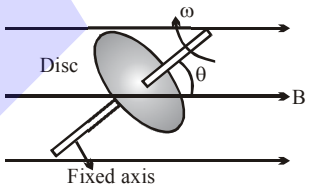
34. The square loop ABCD, carrying a current I, is placed in a uniform magnetic field B, as shown. The loop can rotate about the axis XX'. The plane of the loop makes an angle θ (θ < 90°) with the direction of B. Through what angle will the loop rotate by itself before the torque on it becomes zero–



(A) θ (B) 90° – θ (C) 90° + θ (D) 180° – θ

MG0085

35. A disc of radius r and carrying positive charge q is rotating with an angular speed ω in a uniform magnetic field B about a fixed axis as shown in figure, such that angle made by axis of disc with magnetic field is θ. Torque applied by axis on the disc is



(A) $\frac{q\omega r^2 B\sin\theta}{2}$, clockwise (B) $\frac{q\omega r^2 B\sin\theta}{4}$, anticlockwise

(C) $\frac{q\omega r^2 B\sin\theta}{2}$, anticlockwise (D) $\frac{q\omega r^2 B\sin\theta}{4}$, clockwise

MG0087

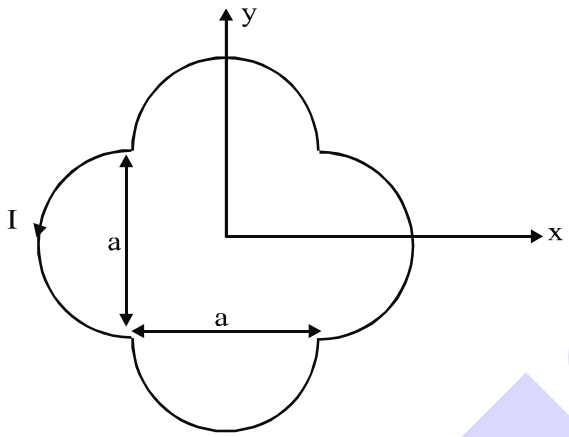
36. A particle of charge q and mass m moves in a circular orbit of radius r with angular speed ω. The ratio of the magnitude of its magnetic moment to that of its angular momentum depends on.

(A) ω and q (B) ω, q and m (C) q and m (D) ω and m

MG0083

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3.Magnetic Effect of Current+EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Ex.p65

37. A loop carrying current 'I' lies in the x-y plane as shown in the figure. The unit vector $\hat{k}$ is coming out of the plane of the paper. The magnetic moment of the current loop is **[JEE 2012]**



(A) $a^2I\hat{k}$ (B) $\left(\frac{\pi}{2}+1\right)a^2I\hat{k}$ (C) $-\left(\frac{\pi}{2}+1\right)a^2I\hat{k}$ (D) $(2\pi+1)a^2I\hat{k}$

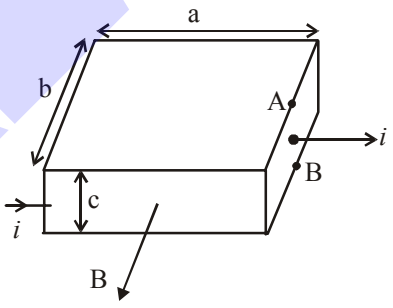
MG0190

38. A moving coil galvanometer has 150 equal divisions. Its current sensitivity is 10 divisions per milli ampere and voltage sensitivity is 2 divisions per millivolt. In order that each division reads 1 V, the resistance in Ohm's needed to be connected in series with the coil will be- **[AIEEE-2005]**

(A) 10^3 (B) 10^5 (C) 99995 (D) 9995

MG0093

39. A current flows through a rectangular conductor in the presence of uniform magnetic field B pointing out of the page as shown. Then the potential difference $V_B - V_A$ is equal to
(Assume charge carries in the conductor to be positively charged moving with speed v)



(A) Bvc (B) –Bvc (C) Bvb (D) –Bvb

MG0092

E. Magnetic materials + Earth magentism

40. A specimen of iron of permeability 8×10^{-3} weber/ amp × metre is placed in a magnetic field of intensity 160 amp/metre. Then magnetic field in this iron is :

(A) 20×10^3 Wb/m^2 (B) 1.28 Wb/m^2 (C) 5×10^{-5} Wb/m^2 (D) 0.8Wb/m^2

MG0108

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current-EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Exp\65

41. Curie temperature is the temperature above which :
(A) a paramagnetic material becomes diamagnetic
(B) a ferromagnetic material becomes diamagnetic
(C) a paramagnetic material becomes ferromagnetic
(D) a ferromagnetic material becomes paramagnetic

MG0109

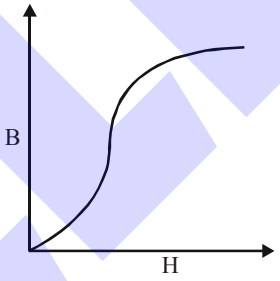
42. If a solution of ferromagnetic material is poured into a U-tube and one arm of this is placed between the poles of a strong magnet with the meniscus in line with the field, then the level of the solution will:
(A) Rise (B) Fall
(C) Rise till the liquid comes out (D) Remain unchanged

MG0110

43. A sample of paramagnetic salt contains 2.0×10^{24} atomic dipoles each of dipole moment 1.5×10^{-23}J T^{-1}. The sample is placed under a homogeneous magnetic field of 0.84T, and cooled to a temperature of 4.2K. The degree of magnetic saturation achieved is equal to 15%. What is the total dipole moment (approximate) of the sample for a magnetic field of 0.98 T and a temperature of 2.8K? (Assume Curie's law).
(A) 7.9 JT^{-1} (B) 52.5 J T^{-1} (C) 30 J T^{-1} (D) 4.6J T^{-1}

MG0111

44. The figure represent B-H curve for commercial iron. As H is increased, permeability :-



(A) increases and becomes constant
(B) increases, reaches a maximum and then decreases
(C) decreases continuously till it becomes very small
(D) decreases reaches a minimum and then increases

MG0115

45. The imagined magnet makes an angle of 11.3° with earth's axis. On the earth, there are two points where magnetic equator and geographic equator meet let the angle of dip and angle of declination at these point be θ & ϕ respectively :-
(A) $\theta = \phi = 0°$ (B) $\theta = 11.3°$, $\phi = 0°$ (C) $\theta = 0°$, $\phi = 11.3°$ (D) $\theta = \phi = 11.3°$

MG0107

46. The total intensity of the earth's magnetic field at the magnetic equator is 5 units. What is its value at a magnetic latitude of 37 degree?
(A) $\sqrt{73}$ units (B) $\sqrt{52}$ units (C) 4 units (D) 3 units

MG0105

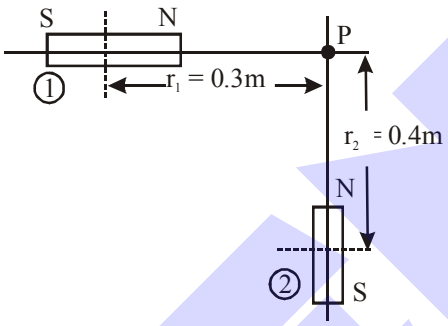
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3.Magnetic Effect of Current+EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Ex.p65

47. A compass needle kept on a horizontal table points in direction 30° west of north. A line is marked showing this direction and compass is now kept in vertical plane containing this line. The needle points in direction 30° below horizontal. If the plane of compass is still vertical but contains true north south direction at what angle to horizontal will needle point :-

(A) $\tan^{-1}\left(\frac{3}{2}\right)$ (B) $\tan^{-1}\left(\frac{2}{3}\right)$ (C) $\tan^{-1}(2)$ (D) $\tan^{-1}\left(\frac{1}{2}\right)$

MG0118

48. Two short magnets of magnetic moment $2Am^2$ and $5Am^2$ respectively are placed along two lines drawn at right angle to each other on the sheet of paper as shown in the figure. What is the magnetic field at the point of intersection of their axis:



(A) 2.15×10^{-5} T (B) 215×10^{-5} T (C) 2.15×10^{-4} T (D) 21.5×10^{-5} T

MG0094

49. A steel wire of length L has a magnetic moment M. It is then bent into a semi-circular arc; the new magnetic moment will be :

(A) M (B) $2M/\pi$ (C) M/L (D) $M \times L$

MG0095

50. The magnetic field at a point on the axis of a dipole is 20 Wb/m^2 Northwards. What will be the magnetic induction at the point if the dipole be rotated anticlockwise by 90° :-

(A) 10 Wb/m^2 Eastwards (B) 10 Web/m^2 Southwards
(C) 10 Wb/m^2 Northwards (D) 20 Wb/m^2 Southwards

MG0100

51. A bar magnet of magnetic moment 2A-m^2 free to rotate about a vertical axis passing through its centre. The magnet is released from rest from east-west position. Then the kinetic energy of the magnet as it takes North-South position is (horizontal component of the earth's field is 25μT and angle of declination is zero :-

(A) 25 μJ (B) 50 μJ (C) 100 μJ (D) 125 μJ

MG0101

52. A magnet is suspended in the magnetic meridian with an untwisted wire. The upper end of wire is rotated through 180° to deflect the magnet by 30° from magnetic meridian. When the magnet is replaced by another magnet, the upper end of wire is rotated through 270° to deflect the magnet 30° from magnetic meridian. The ratio of magnetic moments of magnets is :-

(A) 1 : 5 (B) 1 : 8 (C) 5 : 8 (D) 8 : 5

MG0106

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current-EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Exp\65

MULTIPLE CORRECT TYPE QUESTIONS

A. Biot-Savart law + Ampere's law

53. Consider three quantities x = E/B, y = $\sqrt{1/\mu_0\varepsilon_0}$ and z = $\frac{l}{CR}$. Here, l is the length of a wire, C is a capacitance and R is a resistance. All other symbols have standard meanings.
(A) x, y have the same dimensions
(B) y, z have the same dimensions
(C) z, x have the same dimensions
(D) none of the three pairs have the same dimensions.

MG0124

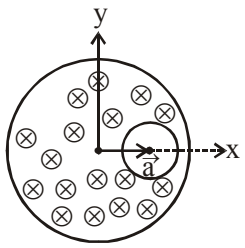
54. The magnetic lines of force due to a straight current carrying wire will be:
(A) circular for finite length of wire (B) circular for semi-infinite wire
(C) circular for infinite wire (D) Parabolic for finite wire

MG0125

55. A long straight wire carries a current along the x-axis. Consider the points A(0, 1, 0), B(0, 1, 1), C(1, 0, 1) and D(1, 1, 1). Which of the following pairs of points will have magnetic fields of the same magnitude?
(A) A and B (B) A and C (C) B and C (D) B and D

MG0131

56. Figure shows cross-section view of a infinite cylindrical wire with a cylinderical cavity, current density is uniform $\vec{J} = -j_0\hat{k}$ as shown in figure.

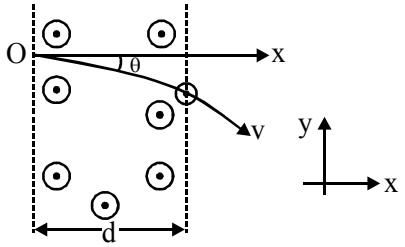


(A) Field inside cavity is uniform
(B) Field inside cavity is along $\vec{a}$
(C) Field inside cavity is perpendicular to $\vec{a}$
(D) If an electron is projected with velocity $v_0\ \hat{j}$ it will move undeviated before colliding with cavity wall

MG0127

B. Magnetic force on a charged particle

57. A particle of charge q is projected with a momentum $\vec{P} = P\hat{i}$ in the given region of magnetic field $\vec{B} = B\hat{k}$. It emerges from the magnetic field after deviating through an angle θ = 30°.
(A) The value of $\vec{P}$ is 2qBd
(B) The value of $\vec{P}$ is qBd



(C) Maximum change in momentum takes place for d ≥ $\frac{P}{qB}$
(D) Maximum change in momentum takes place for d ≥ $\frac{P}{2qB}$

MG0129

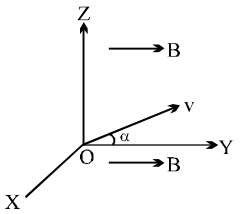
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58. Two identical charged particles enter a uniform magnetic field with same speed but at angles 30° and 60° with field. Let a, b and c be the ratio of their time periods, radii and pitches of the helical paths than

(A) abc = 1 (B) abc > 1 (C) abc < 1 (D) a = bc

MG0132

59. In a region of space, a uniform magnetic field B exists in the y-direction. A proton is fired from the origin, with its initial velocity v making a small angle α with the y-direction in the yz plane. In the subsequent motion of the proton,



(A) its x-coordinate can never be positive

(B) its x- and z-coordinates cannot both be zero at the same time

(C) its z-coordinate can never be negative

(D) its y-coordinate will be proportional to the square of its time elapsed

MG0134

60. An electron is moving along the positive X-axis. You want to apply a magnetic field for a short time so that the electron may reverse its direction and move parallel to the negative X–axis. This can be done by applying the magnetic field along :-

(A) Y-axis (B) Z-axis (C) Y-axis only (D) Z-axis only

MG0133

61. Consider the motion of a positive point charge in a region where there are simultaneous uniform electric and magnetic fields $\vec{E} = E_0\hat{j}$ and $\vec{B} = B_0\hat{j}$. At time t = 0, this charge has velocity $\vec{v}$ in the x-y plane, making an angle θ with the x-axis. Which of the following option (s) is (are) correct for time t > 0? **[IIT-JEE 2012]**

(A) If θ = 0°, the charge moves in a circular path in the x-z plane.

(B) If θ = 0°, the charge undergoes helical motion with constant pitch along the y-axis.

(C) If θ = 10°, the charge undergoes helical motion with its pitch increasing with time, along the y-axis

(D) If θ = 90°, the charge undergoes linear but accelerated motion along the y-axis

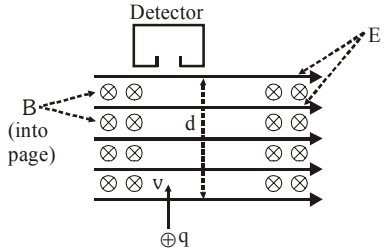
MG0187

COMPREHENSION TYPE QUESTIONS

B. Magnetic force on a charged particle

Paragraph for Questions no. 62 to 65

A velocity filter uses the properties of electric and magnetic fields to select charged particles that are moving with a specific velocity. Charged particles with varying speeds are directed into the filter as shown in figure. The filter consists of an electric field E and a magnetic field B, each of constant magnitude, directed perpendicular to each other as shown. The particles that move straight through the filter with their direction unaltered by the fields have the specific filter speed, v_0. Those with speeds to v_0 may experience sufficiently little deflection that they also enter the detector.



node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current-EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Exp\65

The charged particle will experience a force due to the electric field given by the relationship $\vec{F} = q\vec{E}$, where q is the charge of the particle and $\vec{E}$ is the electric field. The moving particle will also experience a force due to the magnetic field. This force acts to oppose the force due to the electric field. The strength of the force due to the magnetic field is given by the relationship $\vec{F} = q(\vec{v} \times \vec{B})$, where q is the charge of the particle, $\vec{v}$ is the speed of the particle, and $\vec{B}$ is the magnetic field strength. When the forces due to the two fields are equal and opposite, the net force on the particle will be zero, and the particle will pass through the filter with its path unaltered. The electric and magnetic field strengths can be adjusted to choose the specific velocity to be filtered. The effects of gravity can be neglected.

62. The electric and magnetic fields in the filter of figure are adjusted to detect particles with positive charge q of a certain speed, v_0. Which of the following expressions is equal to this speed ?

(A) $B/(q^2E)$ (B) $E/(q^2B)$ (C) B/E (D) E/B

MG0135

63. Which of the following is true about the velocity filter shown in figure ?

(A) It would not work with negatively charged particles

(B) The wider the detector entrance, the more narrow the range of speed detected

(C) The greater the distance d, the more narrow the range of speeds detected

(D) The detector may not detect a charged particle with the desired filter speed if its charge is too high

MG0136

64. Which of the following statements is true regarding a charged particle that is moving through the filter at a speed that is less than the filter speed ?

(A) It experiences a greater force due to the magnetic field than due to the electric field

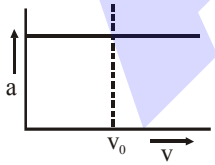
(B) It experiences a greater force due to the electric field than due to the magnetic field

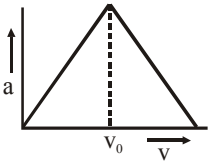
(C) It experiences equal force due to both fields but greater acceleration due to the electric field

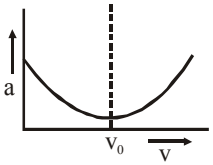
(D) It experiences equal force due to both fields but greater acceleration due to the magnetic field

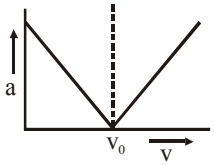
MG0137

65. Particles of identical mass and charge are sent through the filter at varying speeds, and the magnitude of acceleration of each particle is recorded as it first begins to be deflected. If the filter is set to detect particles of speed v_0, which one of the following is correct graph between acceleration and velocity of particle:

(A) 

(B) 

(C) 

(D) 

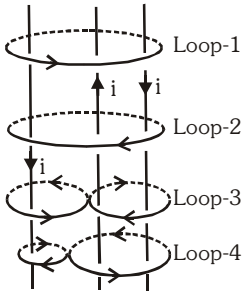
MG0138

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3.Magnetic Effect of Current+EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Ex.p65

MATRIX MATCH TYPE QUESTION

A. Biot-Savart law + Ampere's law

66. Three wires are carrying same constant current i in different directions. Four loops enclosing the wires in different manners are shown. The direction of $d\vec{\ell}$ is shown in figure.



	Column-I		Column-II
(A)	Along closed loop–1	(P)	$\oint \vec{B}.d\vec{\ell} = \mu_0 i$
(B)	Along closed loop–2	(Q)	$\oint \vec{B}.d\vec{\ell} = -\mu_0 i$
(C)	Along closed loop–3	(R)	$\oint \vec{B}.d\vec{\ell} = 0$
(D)	Along closed loop–4	(S)	Net work done by the magnetic force to move a unit charge along the loop is zero

MG0139

D. Torque on a current carrying loop

67.

	Column-I (Magnetic moment of)		Column-II
(A)	a uniformly charged ring rotating uniformly about its axis	(P)	$\frac{q\omega r^2}{5}$
(B)	a charged particle rotating uniformly about a point	(Q)	$\frac{q\omega r^2}{4}$
(C)	a uniformly charged disk rotating uniformly about its axis	(R)	$\frac{q\omega r^2}{3}$
(D)	a uniformly charged spherical shell rotating uniformly about one of its diameter	(S)	$\frac{q\omega r^2}{2}$
(E)	a uniformly charged sphere rotating	(T)	$q\omega r^2$

MG0140

E. Magnetic materials + Earth magentism

68. Match the correct columns :

	Column-I		Column-II
(A)	Isoclinic lines	(P)	Horizontal component of magnetic field is same
(B)	Isogonic lines	(Q)	Angle of declination is same
(C)	Isodynamic lines	(R)	Angle of dip is same

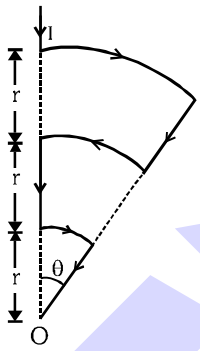
MG0369

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current-EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Exp\65

EXERCISE (S)

A. Biot-Savart law + Ampere's law

1. A conductor carrying a current I is shown in the figure. Calculate the magnetic field intensity at the point O (common centre of all the three arcs).



MG0370

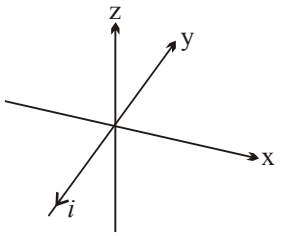
2. Two concentric circular coils X and Y of radii 16 cm and 10 cm, respectively, lie in the same vertical plane containing the north to south direction. Coil X has 20 turns and carries a current of 16A; coil Y has 25 turns and carries a current of 18 A. The sense of the current in X is anticlockwise, and clockwise in Y, for an observer looking at the coils facing west. Give the magnitude and direction of the net magnetic field due to the coils at their centre. ***(NCERT)***

MG0003

3. Two long, straight wires, each carrying a current of 5 A, are placed along the X- and Y-axes respectively. The currents point along the positive directions of the axes. Find the magnetic field at the points (a) (1 m, 1 m), (b) (–1 m, 1 m), (c) (– 1 m, – 1 m) and (d) (1 m, – 1 m).

MG0001

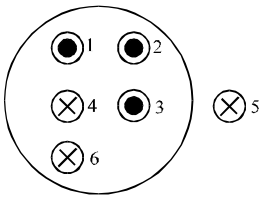
4. A long straight wire carries a current of 10 A directed along the negative y-axis as shown in figure. A uniform magnetic field B_0 of magnitude 10^{-6} T is directed parallel to the x-axis. What is the resultant magnetic field at the following points?
(a) x = 0, z = 2 m; (b) x = 2 m, z = 0; (c) x = 0, z = – 0.5 m



MG0006

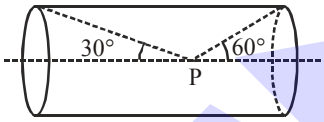
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3.Magnetic Effect of Current+EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Ex.p65

5. Six wires of current $I_1 = 1A, I_2 = 2A, I_3 = 3A, I_4 = 1A, I_5 = 5A$ and $I_6 = 4A$ cut the page perpendicularly at the points 1, 2, 3, 4, 5 and 6 respectively as shown in the figure. Find the value of the integral $\oint \vec{B}.d\vec{\ell}$ around the closed path.



MG0007

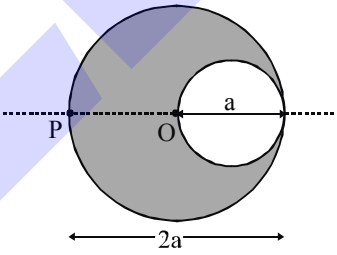
6. Find out magnetic field at axial point 'P' of solenoid shown in figure (where turn density 'n' and current through it is I)



MG0371

7. A cylindrical cavity of diameter a exists inside a cylinder of diameter 2a as shown in the figure. Both the cylinder and the cavity are infinitely long. A uniform current density J flows along the length. If the magnitude of the magnetic field at the point P is given by $\frac{N}{12}\mu_0 aJ$, then the value of N is :

[IIT-JEE 2012]



MG0188

B. Magnetic force on a charged particle

8. A charged particle (charge q, mass m) has velocity v_0 at origin in +x direction. In space there is a uniform magnetic field B in – z direction. Find the y coordinate of particle when is crosses y axis.

MG0008

9. A beam of protons with a velocity 4×10^5 m/s enters a uniform magnetic field of 0.3 T at an angle of 60° to the magnetic field. Find the radius of the helical path taken by the proton beam. Also find the pitch of the helix (which is the distance travelled by a proton in the beam parallel to the magnetic field during one period of rotation).

MG0009

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current-EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Exp\65

C. Magnetic force on a current carrying wire

10. A straight horizontal conducting rod of length 0.45 m and mass 60g is suspended by two vertical wires at its ends. A current of 5.0 A is set up in the rod through the wires. ***(NCERT)***

(a) What magnetic field should be set up normal to the conductor in order that the tension in the wires is zero ?

(b) What will be the total tension in the wires if the direction of current is reversed keeping the magnetic field same as before ? [Ignore the mass of the wires). $g = 9.8\ ms^{-2}$.

MG0011

D. Torque on a current carrying loop

11. A circular coil of 20 turns and radius 10 cm is placed in a uniform magnetic field of 0.10 T normal to the plane of the coil. If the current in the coil is 5.0 A, what is the ***(NCERT)***

(a) total, torque on the coil,

(b) total force on the coil,

(c) average force on each electron in the coil due to the magnetic field ?

(The coil is made of copper wire of cross-sectional area $10^{-5}\ m^2$, and the free electron density in copper is given to be about $10^{29}\ m^{-3}$.)

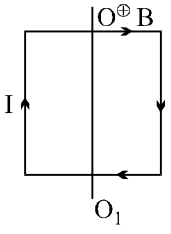
MG0014

12. (a) A circular coil of 30 turns and radius 8.0 cm carrying a current of 6.0 A is suspended vertically in a uniform horizontal magnetic field of magnitude 1.0 T. The field lines make an angle of 60° with the normal of the coil. Calculate the magnitude of the counter torque that must be applied to prevent the coil from turning. ***(NCERT)***

(b) Would your answer change, if the circular coil in (a) were replaced by a planar coil of some irregular shape that encloses the same area? (All other particulars are also unaltered.)

MG0012

13. A square current carrying loop made of thin wire and having a mass m =10g can rotate without friction with respect to the vertical axis OO_1, passing through the centre of the loop at right angles to two opposite sides of the loop. The loop is placed in a homogeneous magnetic field with an induction $B = 10^{-1}$ T directed at right angles to the plane of the drawing. A current I = 2A is flowing in the loop. Find the period of small oscillations that the loop performs about its position of stable equilibrium.



MG0015

14. Two moving coil meters. M_1 and M_2 have the following particulars : ***(NCERT)***

$R_1 = 10\ \Omega$, $N_1 = 30$,

$A_1 = 3.6 \times 10^{-3}\ m^2$. $B_1 = 0.25$ T

$R_2 = 14\ \Omega$, $N_2 = 42$ $A_2 = 1.8 \times 10^{-3}\ m^2$, $B_2 = 0.50$ T

(The spring constants are identical for the two meters). Determine the ratio of (a) current sensitivity and (b) voltage sensitivity of M_2 and M_1.

MG0016

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3.Magnetic Effect of Current+EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Ex.p65

EXERCISE-JM

1. Two short bar magnets of length 1 cm each have magnetic moments 1.20 Am^2 and 1.00 Am^2 respectively. They are placed on a horizontal table parallel to each other with their N poles pointing towards the South. They have a common magnetic equator and are separated by a distance of 20.0 cm. The value of the resultant horizontal magnetic induction at the midpoint O of the line joining their centres is close to :- (Horizontal component of earth's magnetic induction is 3.6×10^{-5} Wb/m^2) **[JEE-Main- 2013]**

(1) 3.6×10^{-5} Wb/m^2 (2) 2.56×10^{-4} Wb/m^2
(3) 3.50×10^{-4} Wb/m^2 (4) 5.80×10^{-4} Wb/m^2

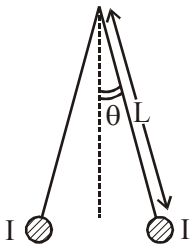
MG0171

2. The coercivity of a small magnet where the ferromagnet gets demagnetized is 3×10^3 A m^{-1}. The current required to be passed in a solenoid of length 10 cm and number of turns 100, so that the magnet gets demagnetized when inside the solenoid, is : **[JEE-Main- 2014]**

(1) 3A (2) 6 A (3) 30 mA (4) 60 mA

MG0173

3. Two long current carrying thin wires, both with current I, are held by the insulating threads of length L and are in equilibrium as shown in the figure, with threads making an angle 'θ' with the vertical. If wires have mass λ per unit length then the value of I is :- (g = gravitational acceleration) **[JEE-Main- 2015]**

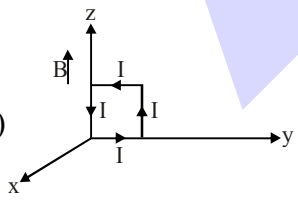


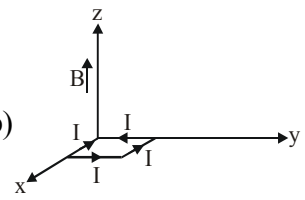
(1) $2\sqrt{\frac{\pi g L}{\mu_0}\tan\theta}$ (2) $\sqrt{\frac{\pi\lambda g L}{\mu_0}\tan\theta}$

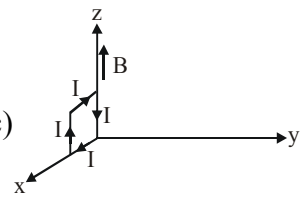
(3) $\sin\theta\sqrt{\frac{\pi\lambda g L}{\mu_0\cos\theta}}$ (4) $2\sin\theta\sqrt{\frac{\pi\lambda g L}{\mu_0\cos\theta}}$

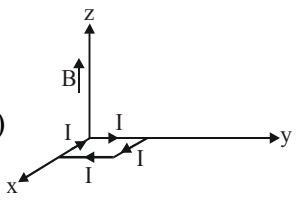
MG0174

4. A rectangular loop of sides 10 cm and 5 cm carrying a current I of 12 A is place in different orientations as shown in the figures below : **[JEE-Main-2015]**

(a) 

(b) 

(c) 

(d) 

If there is a uniform magnetic field of 0.3 T in the positive z direction, in which orientations the loop would be in (i) stable equilibrium and (ii) unstable equilibrium ?

(1) (b) and (d), respectively (2) (b) and (c), respectively
(3) (a) and (b), respectively (4) (a) and (c), respectively

MG0175

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current-EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Exp\65

5. Two coaxial solenoids of different radii carry current I in the same direction. Let $\vec{F}_1$ be the magnetic force on the inner solenoid due to the outer one and $\vec{F}_2$ be the magnetic force on the outer solenoid due to the inner one. Then : **[JEE Main-2015]**

(1) $\vec{F}_1$ is radially inwards and $\vec{F}_2 = 0$

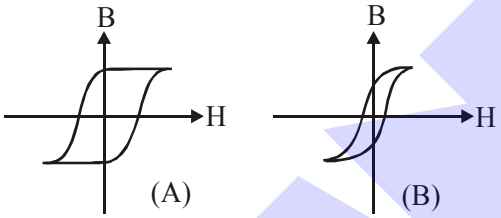
(2) $\vec{F}_1$ is radially outwards and $\vec{F}_2 = 0$

(3) $\vec{F}_1 = \vec{F}_2 = 0$

(4) $\vec{F}_1$ is radially inwards and $\vec{F}_2$ is radially outwards.

MG0176

6. Hysteresis loops for two magnetic materials A and B are given below :



These materials are used to make magnets for electric generators, transformer core and electromagnet core. Then it is proper to use ; **[JEE Main-2016]**

(1) B for electromagnets and transformers.

(2) A for electric generators and transformers.

(3) A for electromagnets and B for electric transformers.

(4) A for transformers and B for electric generators.

MG0177

7. Two identical wires A and B, each of length '*l*', carry the same current I. Wire A is bent into a circle of radius R and wire B is bent to form a square of side 'a'. If B_A and B_B are the values of magnetic field at the centres of the circle and square respectively, then the ratio $\frac{B_A}{B_B}$ is : **[JEE Main-2016]**

(1) $\frac{\pi^2}{8\sqrt{2}}$ (2) $\frac{\pi^2}{8}$ (3) $\frac{\pi^2}{16\sqrt{2}}$ (4) $\frac{\pi^2}{16}$

MG0178

8. A magnetic needle of magnetic moment 6.7×10^{-2} Am^2 and moment of inertia 7.5×10^{-6} kg m^2 is performing simple harmonic oscillations in a magnetic field of 0.01 T. Time taken for 10 complete oscillations is : **[JEE Main-2017]**

(1) 6.98 s (2) 8.76 s (3) 6.65 s (4) 8.89 s

MG0179

9. An electron, a proton and an alpha particle having the same kinetic energy are moving in circular orbits of radii r_e, r_p, r_α respectively in a uniform magnetic field B. The relation between r_e, r_p, r_α is:- **[JEE Main-2018]**

(1) $r_e < r_p = r_\alpha$ (2) $r_e < r_p < r_\alpha$ (3) $r_e < r_\alpha < r_p$ (4) $r_e > r_p = r_\alpha$

MG0180

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3.Magnetic Effect of Current+EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Ex.p65

10. The dipole moment of a circular loop carrying a current I, is m and the magnetic field at the centre of the loop is B_1. When the dipole moment is doubled by keeping the current constant, the magnetic field at the centre of the loop is B_2. The ratio $\frac{B_1}{B_2}$ is : **[JEE Main-2018]**

(1) $\sqrt{3}$ (2) $\sqrt{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) 2

MG0181

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

11. A bar magnet is demagnetized by inserting it inside a solenoid of length 0.2 m, 100 turas, and carrying a current of 5.2 A. The coercivity of the bar magnet is : **[JEE-Main-2019_Jan]**

(1) 1200 A/m (2) 2600 A/m (3) 520 A/jm (4) 285 A/m

MG0227

12. At some location on earth the horizontal component of earth's magnetic field is 18×10^{-6} T. At this location, magnetic neeedle of length 0.12 m and pole strength 1.8 Am is suspended from its mid-point using a thread, it makes 45° angle with horizontal in equilibrium. To keep this needle horizontal, the vertical force that should be applied at one of its ends is : **[JEE-Main-2019_Jan]**

(1) 3.6×10^{-5} N (2) 6.5×10^{-5} N (3) 1.3×10^{-5} N (4) 1.8×10^{-5} N

MG0228

13. An insulating thin rod of length ℓ has a x linear charge density $p(x) = \rho_0 \frac{x}{\ell}$ on it. The rod is rotated about an axis passing through the origin (x = 0) and perpendicular to the rod. If the rod makes n rotations per second, then the time averaged magnetic moment of the rod is : **[JEE-Main-2019_Jan]**

(1) $\frac{\pi}{4}n\rho\ell^3$ (2) $n\rho\ell^3$ (3) $\pi n\rho\ell^3$ (4) $\frac{\pi}{3}n\rho\ell^3$

MG0229

14. A paramagnetic substance in the form of a cube with sides 1 cm has a magnetic dipole moment of 20×10^{-6} J/T when a magnetic intensity of 60×10^3 A/m is applied. Its magnetic susceptibility is :- **[JEE-Main-2019_Jan]**

(1) 2.3×10^{-2} (2) 3.3×10^{-2} (3) 3.3×10^{-4} (4) 4.3×10^{-2}

MG0230

15. A galvanometer having a resistance of 20 Ω and 30 divisions on both sides has figure of merit 0.005 ampere/division. The resistance that should be connected in series such that it can be used as a voltmeter upto 15 volt, is :- **[JEE-Main-2019_Jan]**

(1) 80 Ω (2) 120 Ω (3) 125 Ω (4) 100 Ω

MG0231

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current-EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Exp\65

16. A particle of mass m and charge q is in an electric and magnetic field given by

$\vec{E} = 2\hat{i} + 3\hat{j}$; $\vec{B} = 4\hat{j} + 6\hat{k}$.

The charged particle is shifted from the origin to the point P(x = 1 ; y = 1) along a straight path. The magnitude of the total work done is :- **[JEE-Main-2019_Jan]**

(1) (0.35)q (2) (0.15)q (3) (2.5)q (4) 5q

MG0232

17. There are two long co-axial solenoids of same length *l*. the inner and outer coils have radii r_1 and r_2 and number of turns per unit length n_1 and n_2 respectively. The rate of mutual inductance to the self-inductance of the inner-coil is : **[JEE-Main-2019_Jan]**

(1) $\frac{n_2}{n_1}\cdot\frac{r_2^2}{r_1^2}$ (2) $\frac{n_2}{n_1}\cdot\frac{r_1}{r_2}$ (3) $\frac{n_1}{n_2}$ (4) $\frac{n_2}{n_1}$

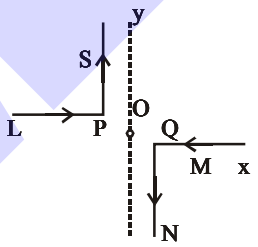
MG0233

18. A paramagnetic material has 10^{28}atoms/m^3. Its magnetic susceptibility at temperature 350 K is 2.8×10^{-4}. Its susceptibility at 300 K is : **[JEE-Main-2019_Jan]**

(1) 3.672×10^{-4} (2) 3.726×10^{-4} (3) 3.267×10^{-4} (4) 2.672×10^{-4}

MG0234

19. As shown in the figure, two infinitely long, identical wires are bent by 90° and placed in such a way that the segments LP and QM are along the x-axis, while segments PS and QN are parallel to the y-axis. If OP = OQ = 4cm, and the magnitude of the magnetic field at O is 10^{-4} T, and the two wires carry equal currents (see figure), the magnitude of the current in each wire and the direction of the magnetic field at O will be ($\mu_0 = 4\pi \times 10^{-7}$ NA^{-2}) : **[JEE-Main-2019_Jan]**



(1) 40 A, perpendicular into the page (2) 40 A, perpendicular out of the page
(3) 20 A, perpendicular out of the page (4) 20 A, perpendicular into the page

MG0235

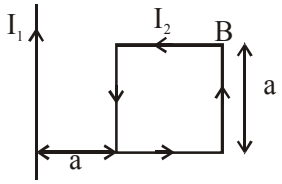
20. Two coils 'P' and 'Q' are separated by some distance. When a current of 3 A flows through coil 'P', a magnetic flux of 10^{-3} Wb passes through 'Q'. No current is passed through 'Q'. When no current passes through 'P' and a current of 2 A passes through 'Q', the flux through 'P' is :- **[JEE-Main-2019_April]**

(1) 6.67×10^{-3} Wb (2) 6.67×10^{-4} Wb
(3) 3.67×10^{-4} Wb (4) 3.67×10^{-3} Wb

MG0237

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3.Magnetic Effect of Current+EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Exp\65

21. A rigid square loop of side 'a' and carrying current I_2 is lying on a horizontal surface near a long current I_1 carrying wire in the same plane as shown in figure. The net force on the loop due to wire will be : **[JEE-Main-2019_April]**



(1) Attractive and equal to $\frac{\mu_0 I_1 I_2}{3\pi}$

(2) Repulsive and equal to $\frac{\mu_0 I_1 I_2}{4\pi}$

(3) Repulsive and equal to $\frac{\mu_0 I_1 I_2}{2\pi}$

(4) Zero

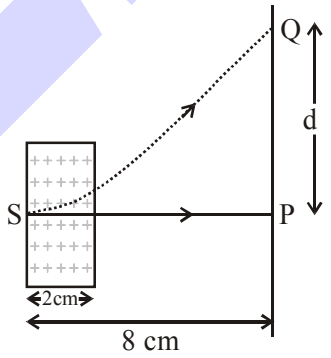
MG0238

22. A rectangular coil (Dimension 5 cm × 2.5 cm) with 100 turns, carrying a current of 3 A in the clock-wise direction is kept centered at the origin and in the X-Z plane. A magnetic field of 1 T is applied along X-axis. If the coil is tilted through 45° about Z-axis, then the torque on the coil is : **[JEE-Main-2019_April]**

(1) 0.55 Nm (2) 0.27 Nm (3) 0.38 Nm (4) 0.42 Nm

MG0239

23. An electron, moving along the x-axis with an initial energy of 100 eV, enters a region of magnetic field $\vec{B} = (1.5\times10^{-3}\,T)\hat{k}$ at S (See figure). The field extends between x = 0 and x = 2 cm. The electron is detected at the point Q on a screen placed 8 cm away from the point S. The distance d between P and Q (on the screen) is :
(electron's charge = 1.6×10^{-19} C, mass of electron = 9.1×10^{-31} kg) **[JEE-Main-2019_April]**



(1) 12.87 cm (2) 1.22 cm (3) 11.65 cm (4) 2.25 cm

MG0240

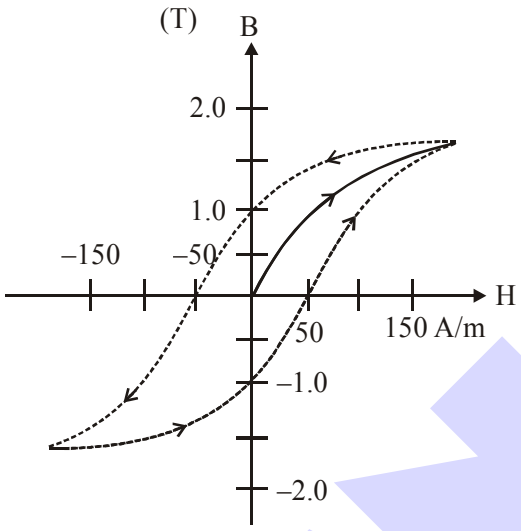
24. A magnetic compass needle oscillates 30 times per minute at a place where the dip is 45°, and 40 times per minute where the dip is 30°. If B_1 and B_2 are respectively the total magnetic field due to the earth at the two places, then the ratio B_1/B_2 is best given by : **[JEE-Main-2019_April]**

(1) 2.2 (2) 1.8 (3) 0.7 (4) 3.6

MG0241

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3\Magnetic Effect of Current-EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Exp\65

25. The figure gives experimentally measured B vs. H variation in a ferromagnetic material. The retentivity, co-ercivity and saturation, respectively, of the material are: **[JEE-Main-2020_Jan.]**



(1) 150 A/m, 1.0 T and 1.5 T
(2) 1.0 T, 50 A/m and 1.5 T
(3) 1.5 T, 50 A/m and 1.0 T
(4) 1.5 T, 50 A/m and 1.0 T

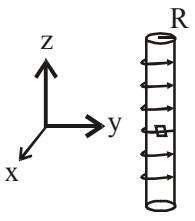
MG0242

26. Consider a circular coil of wire carrying constant current I, forming a magnetic dipole. The magnetic flux through an infinite plane that contains the circular coil and excluding the circular coil area is given by ϕ_i. The magnetic flux through the area of the circular coil area is given by ϕ_0. Which of the following option is correct ? **[JEE-Main-2020_Jan.]**

(1) $\phi_i = -\phi_0$ (2) $\phi_i = \phi_0$ (3) $\phi_i < \phi_0$ (4) $\phi_i > \phi_0$

MG0243

27. An electron gun is placed inside a long solenoid of radius R on its axis. The solenoid has n turns/length and carries a current I. The electron gun shoots an electron along the radius of the solenoid with speed v. If the electron does not hit the surface of the solenoid, maximum possible value of v is (all symbols have their standard meaning) : **[JEE-Main-2020_Jan.]**



(1) $\frac{e\mu_0 nIR}{m}$ (2) $\frac{e\mu_0 nIR}{2m}$ (3) $\frac{2e\mu_0 nIR}{m}$ (4) $\frac{e\mu_0 nIR}{4m}$

MG0244

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3.Magnetic Effect of Current+EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Ex.p65

28. Magnetic materials used for making permanent magnets (P) and magnets in a transformer (T) have different properties of the following, which property best matches for the type of magnet required? **[JEE-Main-2020_Sep.]**

(1) T : Large retentivity, small coercivity
(2) P : Small retentivity, large coercivity
(3) T : Large retentivity, large coercivity
(4) P : Large retentivity, large coercivity

MG0245

29. A galvanometer coil has 500 turns and each turn has an average area of 3×10^{-4} m^2. If a torque of 1.5Nm is required to keep this coil parallel to magnetic field when a current of 0.5 A is flowing through it, the strength of the field (in T) is _____ . **[JEE-Main-2020_Sep.]**

MG0247

30. A paramagnetic sample shows a net magnetisation of 6 A/m when it is placed in an external magnetic field of 0.4 T at a temperature of 4 K. When the sample is placed in an external magnetic field of 0.3 T at a temperature of 24 K, then the magnetisation will be : **[JEE-Main-2020_Sep.]**

(1) 4 A/m (2) 0.75 A/m (3) 2.25 A/m (4) 1 A/m

MG0248

31. An iron rod of volume 10^{-3} m^3 and relative permeability 1000 is placed as core in a solenoid with 10 turns/cm. If a current of 0.5 A is passed through the solenoid, then the magnetic moment of the rod will be : **[JEE-Main-2020_Sep.]**

(1) 0.5×10^2 Am^2 (2) 50×10^2 Am^2 (3) 500×10^2 Am^2 (4) 5×10^2 Am^2

MG0249

32. A square loop of side 2a and carrying current I is kept in xz plane with its centre at origin. A long wire carrying the same current I is placed parallel to z-axis and passing through point (0, b, 0), (b >> a). The magnitude of torque on the loop about z-ax is will be : **[JEE-Main-2020_Sep.]**

(1) $\frac{2\mu_0 I^2 a^2 b}{\pi(a^2+b^2)}$ (2) $\frac{\mu_0 I^2 a^2 b}{2\pi(a^2+b^2)}$ (3) $\frac{\mu_0 I^2 a^2}{2\pi b}$ (4) $\frac{2\mu_0 I^2 a^2}{\pi b}$

MG0251

33. A soft ferromagnetic material is placed in an external magnetic field. The magnetic domains : **[JEE-Main-2021_Feb.]**

(1) increase in size but no change in orientation.
(2) have no relation with external magnetic field.
(3) decrease in size and changes orientation.
(4) may increase or decrease in size and change its orientation.

MG0372

34. In a ferromagnetic material, below the curie temperature, a domain is defined as : **[JEE-Main-2021_Feb.]**

(1) a macroscopic region with zero magnetization.
(2) a macroscopic region with consecutive magnetic dipoles oriented in opposite direction.
(3) a macroscopic region with randomly oriented magnetic dipoles.
(4) a macroscopic region with saturation magnetization.

MG0373

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current-EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Exp\65

35. Which of the following statements are correct? **[JEE-Main-2021_March]**

(A) Electric monopoles do not exist whereas magnetic monopoles exist.

(B) Magnetic field lines due to a solenoid at its ends and outside cannot be completely straight and confined.

(C) Magnetic field lines are completely confined within a toroid.

(D) Magnetic field lines inside a bar magnet are not parallel.

(E) $\chi = -1$ is the condition for a perfect diamagnetic material, where χ is its magnetic susceptibility.

Choose the correct answer from the options given below :

(1) (C) and (E) only (2) (B) and (D) only (3) (A) and (B) only (4) (B) and (C) only

MG0374

36. **Statement I :** The ferromagnetic property depends on temperature. At high temperature, ferromagnet becomes paramagnet.

Statement II : At high temperature, the domain wall area of a ferromagnetic substance increases.

In the light of the above statements, choose the **most appropriate** answer from the options given below: **[JEE-Main-2021_July]**

(1) **Statement I** is true but **Statement II** is false

(2) Both **Statement I** and **Statement II** are true

(3) Both **Statement I** and **Statement II** are false

(4) **Statement I** is false but **Statement II** is true

MG0375

37. Match List I with List II.

	List-I		**List-II**
(a)	Capacitance, C	(i)	$M^1L^1T^{-3}A^{-1}$
(b)	Permittivity of free space, ε_0	(ii)	$M^{-1}L^{-3}T^4A^2$
(c)	Permeability of free space, μ_0	(iii)	$M^{-1}L^{-2}T^4A^2$
(d)	Electric field, E	(iv)	$M^1L^1T^{-2}A^{-2}$

Choose the correct answer from the options given below **[JEE-Main-2021_July]**

(1) (a) → (iii), (b) → (ii), (c) → (iv), (d) → (i)

(2) (a) → (iii), (b) → (iv), (c) → (ii), (d) → (i)

(3) (a) → (iv), (b) → (ii), (c) → (iii), (d) → (i)

(4) (a) → (iv), (b) → (iii), (c) → (ii), (d) → (i)

MG0376

38. If the maximum value of accelerating potential provided by a ratio frequency oscillator is 12 kV. The number of revolution made by a proton in a cyclotron to achieve one sixth of the speed of light is ..... [$m_p = 1.67 \times 10^{-27}$ kg, $e = 1.6 \times 10^{-19}$ C, Speed of light = 3×10^8 m/s]

[JEE-Main-2021_Aug.]

MG0377

39. A coaxial cable consists of an inner wire of radius 'a' surrounded by an outer shell of inner and outer radii 'b' and 'c' respectively. The inner wire carries an electric current i_0, which is distributed uniformly across cross-sectional area. The outer shell carries an equal current in opposite direction and distributed uniformly. What will be the ratio of the magnetic field at a distance x from the axis when (i) $x < a$ and (ii) $a < x < b$?

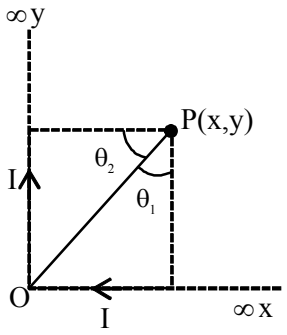
[JEE-Main-2021_Aug.]

(1) $\frac{x^2}{a^2}$ (2) $\frac{a^2}{x^2}$ (3) $\frac{x^2}{b^2 - a^2}$ (4) $\frac{b^2 - a^2}{x^2}$

MG0378

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3.Magnetic Effect of Current+EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Ex.p65

40. There are two infinitely long straight current carrying conductors and they are held at right angles to each other so that their common ends meet at the origin as shown in the figure given below. The ratio of current in both conductor is 1 : 1. The magnetic field at point P is ____. **[JEE-Main-2021_Aug.]**



(1) $\frac{\mu_0 I}{4\pi xy}\left[\sqrt{x^2+y^2}+(x+y)\right]$

(2) $\frac{\mu_0 I}{4\pi xy}\left[\sqrt{x^2+y^2}-(x+y)\right]$

(3) $\frac{\mu_0 Ixy}{4\pi}\left[\sqrt{x^2+y^2}-(x+y)\right]$

(4) $\frac{\mu_0 Ixy}{4\pi}\left[\sqrt{x^2+y^2}+(x+y)\right]$

MG0379

41. Given below are two statements:

Statement - I : Susceptibilities of paramagnetic and ferromagnetic substances increase with decrease in temperature.

Statement - II: Diamagnetism is a result of orbital motions of electrons developing magnetic moments opposite to the applied magnetic field.

Choose the CORRECT answer from the options given below : - **[JEE-Main-2022_June]**

(A) Both statement - I and statement -II are true.

(B) Both statement - I and Statement - II are false.

(C) Statement - I is true but statement - II is false.

(D) Statement-I is false but Statement-II is true.

MG0380

42. **Assertion (A) :** Non-polar materials do not have my permanent dipole moment.

Reason (R) : When an non-polar material is placed in a electric field. the centre of the positive charge distribution of it's individual atom or molecule coinsides with the centre of the negative charge distribution.

In the light of above statements, choose the most appropriate answer from the options given below.

[JEE-Main-2022_June]

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(B) Both (A) and (R) are correct and (R) is not the correct explanation of (A).

(C) (A) is correct but (R) is not correct.

(D) (A) is not correct but (R) is correct.

MG0381

node06\B0BC-BD\Kota\JEE\Advanced\Enthuse\Phy\Module\3-Magnetic Effect of Current-EM1 & AC\Eng\01_Magnetic Effect of Current\01_Th + Exp\65

43. Given below are two statements :
Statement I: The electric force changes the speed of the charged particle and hence changes its kinetic energy: whereas the magnetic force does not change the kinetic energy of the charged particle.
Statement II: The electric force accelerates the positively charged particle perpendicular to the direction of electric field. The magnetic force accelerates the moving charged particle along the direction of magnetic field. In the light of the above statements, choose the most appropriate answer from the options given below: **[JEE-Main-2022_June]**
(A) Both Statement I and Statement II are correct.
(B) Both Statement I and Statement II are incorrect.
(C) Statement I is correct but Statement II is incorrect.
(D) Statement I is incorrect but Statement II is correct.

MG0382

44. A velocity selector consists of electric field $\vec{E} = E\hat{k}$ and magnetic field $\vec{B} = B\hat{j}$ with B = 12 mT. The value E required for an electron of energy 728 eV moving along the positive x-axis to pass undeflected is : (Given, mass of electron = 9.1×10^{-31} kg) **[JEE-Main-2022_July]**
(A) 192 kVm^{-1} (B) 192 m Vm^{-1} (C) 9600 kVm^{-1} (D) 16 kVm^{-1}

MG0383

45. Two bar magnets oscillate in a horizontal plane in earth's magnetic field with time periods of 3 s and 4 s respectively. If their moments of inertia are in the ratio of 3 : 2 then the ratio of their magnetic moments will be : **[JEE-Main-2022_July]**
(A) 2 : 1 (B) 8 : 3 (C) 1 : 3 (D) 27 : 16

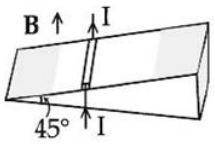
MG0384

46. A magnet hung at 45º with magnetic meridian makes an angle of 60º with the horizontal. The actual value of the angle of dip is **[JEE-Main-2022_July]**

(A) $\tan^{-1}\left(\sqrt{\frac{3}{2}}\right)$ (B) $\tan^{-1}\left(\sqrt{6}\right)$ (C) $\tan^{-1}\left(\sqrt{\frac{2}{3}}\right)$ (D) $\tan^{-1}\left(\sqrt{\frac{1}{2}}\right)$

MG0385

47. As shown in the figure, a metallic rod of linear density 0.45 kg m^{-1} is lying horizontally on a smooth incline plane which makes an angle of 45° with the horizontal. The minimum current flowing in the rod required to keep it stationary, when 0.15 T magnetic field is acting on it in the vertical upward direction, will be : [Use $g = 10\ m/s^2$] **[JEE-Main-2022_July]**



(A) 30 A (B) 15 A (C) 10 A (D) 3 A

MG0386

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3.Magnetic Effect of Current+EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Ex.p65

EXERCISE-JA

1. A particle of mass M and positive charge Q, moving with a constant velocity $\vec{u}_1 = 4\hat{i}\,ms^{-1}$, enters a region of uniform static magnetic field normal to the x-y plane. The region of the magnetic field extends from x = 0 to x = L from all values of y. After passing through this region, the particle emerges on the other side after 10 milliseconds with a velocity $\vec{u}_2 = 2\left(\sqrt{3}\hat{i} + \hat{j}\right)ms^{-1}$. The correct statements(s) is (are) :- **[IIT-JEE 2013]**

(A) The direction of the magnetic field is –z direction.

(B) The direction of the magnetic field is +z direction.

(C) The magnitude of the magnetic field $\frac{50\pi M}{3Q}$ units.

(D) The magnitude of the magnetic field is $\frac{100\pi M}{3Q}$ units.

MG0191

2. A steady current I flows along an infinitely long hollow cylindrical conductor of radius R. This cylinder is placed coaxially inside an infinite solenoid of radius 2R. The solenoid has n turns per unit length and carries a steady current I. Consider a point P at a distance r from the common axis. The correct statement(s) is (are) :- **[JEE-Advanced-2013]**

(A) In the region 0 < r < R, the magnetic field is non-zero

(B) In the region R < r < 2R, the magnetic field is along the common axis

(C) In the region R < r < 2R, the magnetic field is tangential to the circle of radius r, centred on the axis

(D) In the region r > 2R, the magnetic field is non-zero

MG0192

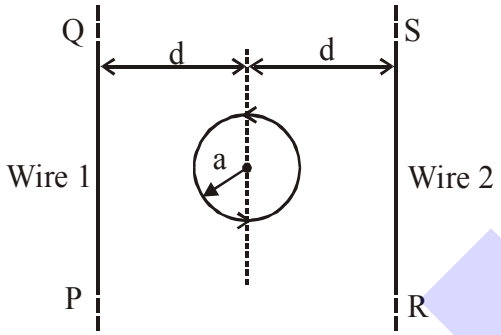
3. Two parallel wires in the plane of the paper are distance X_0 apart. A point charge is moving with speed u between the wires in the same plane at a distance X_1 from one of the wires. When the wires carry current of magnitude I in the same direction, the radius of curvature of the path of the point charge is R_1. In contrast, if the currents I in the two wires have directions opposite to each other, the radius of curvature of the path is R_2. If $\frac{X_0}{X_1} = 3$, and value of $\frac{R_1}{R_2}$ is:- **[JEE-Advanced-2014]**

MG0193

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current-EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Exp\65

Paragraph for Questions 4 & 5

The figure shows a circular loop of radius a with two long parallel wires (numbered 1 and 2) all in the plane of the paper. The distance of each wire from the centre of the loop is d. The loop and the wires are carrying the same current I. The current in the loop is in the counterclockwise direction if seen from above. **[JEE-Advanced-2014]**



4. When d ≈ a but wires are not touching the loop, it is found that the net magnetic field on the axis of the loop is zero at a height h above the loop. In that case

(A) current in wire 1 and wire 2 is the direction PQ and RS, respectively and h ≈ a

(B) current in wire 1 and wire 2 is the direction PQ and SR, respectively and h ≈ a

(C) current in wire 1 and wire 2 is the direction PQ and SR, respectively and h ≈ 1.2 a

(D) current in wire 1 and wire 2 is the direction PQ and RS, respectively and h ≈ 1.2 a

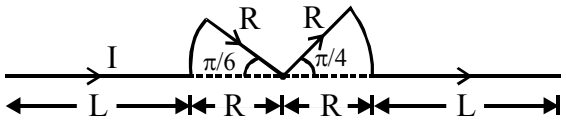
MG0194

5. Consider d >> a, and the loop is rotated about its diameter parallel to the wires by 30° from the position shown in the figure. If the currents in the wires are in the opposite directions, the torque on the loop at its new position will be (assume that the net field due to the wires is constant over the loop) **[JEE-Advanced-2014]**

(A) $\frac{\mu_0 I^2 a^2}{d}$ (B) $\frac{\mu_0 I^2 a^2}{2d}$ (C) $\frac{\sqrt{3}\mu_0 I^2 a^2}{d}$ (D) $\frac{\sqrt{3}\mu_0 I^2 a^2}{2d}$

MG0195

6. A conductor (shown in the figure) carrying constant current I is kept in the x-y plane in a uniform magnetic field $\vec{B}$. If F is the magnitude of the total magnetic force acting on the conductor, then the correct statement(s) is (are) **[JEE-Advanced-2015]**



(A) If $\vec{B}$ is along $\hat{z}$, F ∝ (L + R)

(B) If $\vec{B}$ is along $\hat{x}$, F = 0

(C) If $\vec{B}$ is along $\hat{y}$, F ∝ (L + R)

(D) If $\vec{B}$ is along $\hat{z}$, F = 0

MG0196

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3.Magnetic Effect of Current+EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Ex.p65

Answer Q.7, Q.8 and Q.9 by appropriately matching the information given in the three columns of the following table.

A charged particle (electron or proton) is introduced at the origin (x = 0, y = 0, z = 0) with a given initial velocity $\vec{v}$. A uniform electric field $\vec{E}$ and a uniform magnetic field $\vec{B}$ exist everywhere. The velocity $\vec{v}$, electric field $\vec{E}$ and magnetic field $\vec{B}$ are given in column 1, 2 and 3, respectively. The quantities E_0, B_0 are positive in magnitude. **[JEE-Advanced-2017]**

Column-1	**Column-2**	**Column-3**
(I) Electron with $\vec{v} = 2\frac{E_0}{B_0}\hat{x}$	(i) $\vec{E} = E_0\hat{z}$	(P) $\vec{B} = -B_0\hat{x}$
(II) Electron with $\vec{v} = \frac{E_0}{B_0}\hat{y}$	(ii) $\vec{E} = -E_0\hat{y}$	(Q) $\vec{B} = B_0\hat{x}$
(III) Proton with $\vec{v} = 0$	(iii) $\vec{E} = -E_0\hat{x}$	(R) $\vec{B} = B_0\hat{y}$
(IV) Proton with $\vec{v} = 2\frac{E_0}{B_0}\hat{x}$	(iv) $\vec{E} = E_0\hat{x}$	(S) $\vec{B} = B_0\hat{z}$

7. In which case will the particle move in a straight line with constant velocity ?

(A) (II) (iii) (S) (B) (IV) (i) (S) (C) (III) (ii) (R) (D) (III) (iii) (P)

MG0197

8. In which case will the particle describe a helical path with axis along the positive z-direction ?

(A) (II) (ii) (R) (B) (IV) (ii) (R) (C) (IV) (i) (S) (D) (III) (iii) (P)

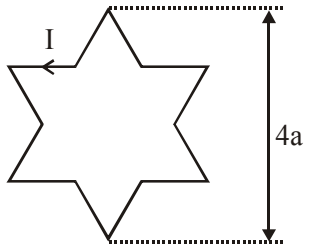
MG0198

9. In which case would the particle move in a straight line along the negative direction of y-axis (i.e., move along $-\hat{y}$) ?

(A) (IV) (ii) (S) (B) (III) (ii) (P) (C) (II) (iii) (Q) (D) (III) (ii) (R)

MG0199

10. A symmetric star shaped conducting wire loop is carrying a steady state current I as shown in the figure. The distance between the diametrically opposite vertices of the star is 4a. The magnitude of the magnetic field at the center of the loop is : **[JEE-Advanced-2017]**

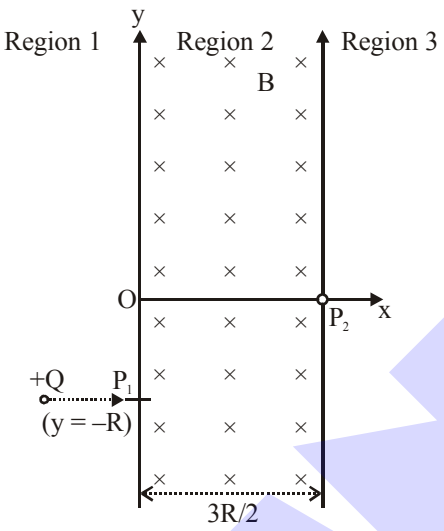


(A) $\frac{\mu_0 I}{4\pi a}3\left[\sqrt{3}-1\right]$ (B) $\frac{\mu_0 I}{4\pi a}6\left[\sqrt{3}-1\right]$ (C) $\frac{\mu_0 I}{4\pi a}6\left[\sqrt{3}+1\right]$ (D) $\frac{\mu_0 I}{4\pi a}3\left[2-\sqrt{3}\right]$

MG0200

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3-Magnetic Effect of Current-EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Exp\65

11. A uniform magnetic field B exists in the region between x = 0 and x = $\frac{3R}{2}$ (region 2 in the figure) pointing normally into the plane of the paper. A particle with charge +Q and momentum p directed along x-axis enters region 2 from region 1 at point $P_1(y = -R)$. Which of the following options(s) is/ are correct ? **[JEE-Advanced-2017]**



(A) For $B = \frac{8}{13}\frac{p}{QR}$, the particle will enter region 3 through the point P_2 on x-axis

(B) For $B > \frac{2}{3}\frac{p}{QR}$, the particle will re-enter region 1

(C) For a fixed B, particle of same charge Q and same velocity v, the distance between the point P_1 and the point of re-entry into region 1 is inversely proportional to the mass of the particle.

(D) When the particle re-enters region 1 through the longest possible path in region 2, the magnitude of the chage in its linear momentum between point P_1 and the farthest point from y-axis is $\frac{p}{\sqrt{2}}$.

MG0201

12. Two infinitely long straight wires lie in the xy-plane along the lines x = ±R. The wire located at x = +R carries a constant current I_1 and the wire located at x = –R carries a constant current I_2. A circular loop of radius R is suspended with its centre at (0, 0, $\sqrt{3}R$) and in a plane parallel to the xy-plane. This loop carries a constant current I in the clockwise direction as seen from above the loop. The current in the wire is taken to be positive if it is in the $+\hat{j}$ direction. Which of the following statements regarding the magnetic field $\vec{B}$ is (are) true ? **[JEE-Advanced-2018]**

(A) If $I_1 = I_2$, then $\vec{B}$ cannot be equal to zero at the origin (0, 0, 0)

(B) If $I_1 > 0$ and $I_2 < 0$, then $\vec{B}$ can be equal to zero at the origin (0, 0, 0)

(C) If $I_1 < 0$ and $I_2 > 0$, then $\vec{B}$ can be equal to zero at the origin (0, 0, 0)

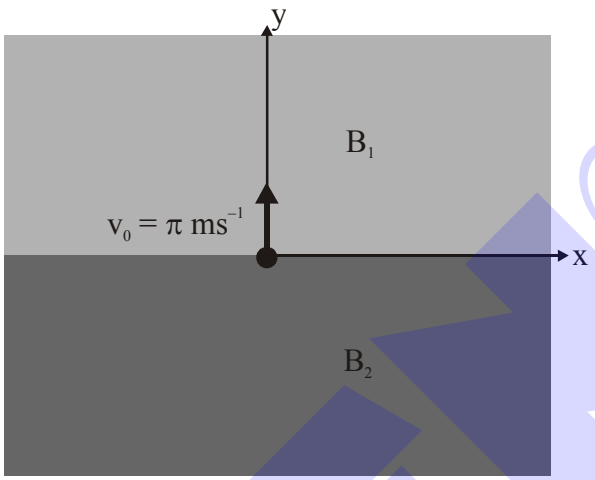
(D) If $I_1 = I_2$, then the z-component of the magnetic field at the centre of the loop is $\left(-\frac{\mu_0 I}{2R}\right)$

MG0202

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\3.Magnetic Effect of Current+EMI & AC\Eng\01_Magnetic Effect of Current\01_Th + Ex.p65

13. In the x-y-plane, the region y > 0 has a uniform magnetic field $B_1\hat{k}$ and the region y < 0 has a another uniform magnetic field $B_2\hat{k}$. A positively charged particle is projected from the origin along the positive y-axis with speed $v_0 = \pi ms^{-1}$ at t = 0, as shown in the figure. Neglect gravity in this problem. Let t = T be the time when the particle crosses the x-axis from below for the first time. If $B_2 = 4B_1$, the average speed of the particle, in ms^{-1}, along the x-axis in the time interval T is _____.

[JEE-Advanced-2018]

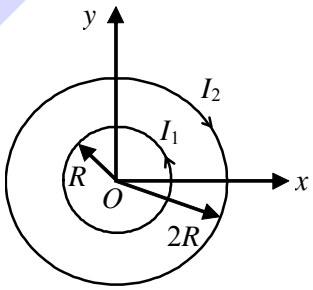


MG0203

14. An α-particle (mass 4 amu) and a singly charged sulfur ion (mass 32 amu) are initially at rest. They are accelerated through a potential V and then allowed to pass into a region of uniform magnetic field which is normal to the velocities of the particles. Within this region, the α-particle and the sulfur ion move in circular orbits of radii r_α and r_s, respectively. The ratio (r_s/r_α) is ___. **[JEE-Advanced-2021]**

MG0252

15. Two concentric circular loops, one of radius R and the other of radius 2R, lie in the xy-plane with the origin as their common center, as shown in the figure. The smaller loop carries current I_1 in the anti-clockwise direction and the larger loop carries current I_2 in the clockwise direction, with $I_2 > 2I_1$. $\vec{B}$ (x, y) denotes the magnetic field at a point (x, y) in the xy-plane. Which of the following statement(s) is(are) current? **[JEE-Advanced-2021]**



(A) $\vec{B}(x,y)$ is perpendicular to the xy-plane at any point in the plane.

(B) $|\vec{B}(x,y)|$ depends on x and y only through the radial distance $r = \sqrt{x^2+y^2}$.

(C) $|\vec{B}(x,y)|$ is non-zero at all points for r < R.

(D) $\vec{B}(x,y)$ points normally outward from the xy-plane for all the points between the two loops

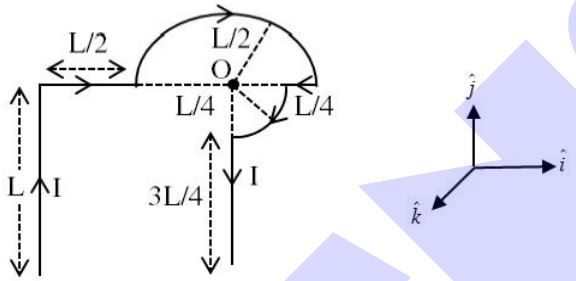
MG0253

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16. In a particular system of units, a physical quantity can be expressed in terms of the electric charge e, electron mass m_e, Planck's constant h, and Coulomb's constant $k = \frac{1}{4\pi \in_0}$, where $\in_0$ is the permittivity of vacuum. In terms of these physical constants, the dimension of the magnetic field is $[B] = [e]^{\alpha} [m_e]^{\beta} [h]^{\gamma} [k]^{\delta}$. The value of $\alpha + \beta + \gamma + \delta$ is _____. **[JEE-Advance-2022]**

MG0387

17. Which one of the following options represents the magnetic field $\vec{B}$ at O due to the current flowing in the given wire segments lying on the xy plane? **[JEE-Advance-2022]**



(A) $\vec{B} = \frac{-\mu_0 I}{L}\left(\frac{3}{2} + \frac{1}{4\sqrt{2}\pi}\right)\hat{k}$

(B) $\vec{B} = -\frac{\mu_0 I}{L}\left(\frac{3}{2} + \frac{1}{2\sqrt{2}\pi}\right)\hat{k}$

(C) $\vec{B} = \frac{-\mu_0 I}{L}\left(1 + \frac{1}{4\sqrt{2}\pi}\right)\hat{k}$

(D) $\vec{B} = \frac{-\mu_0 I}{L}\left(1 + \frac{1}{4\pi}\right)\hat{k}$

MG0388

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ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	B	A	B	D	C	A	B	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	B	A	B	A	D	C	C	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	D	C	B	B	B	A	A	D	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	B	A	D	A	D	C	A	C	C	A
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	B	D	C	A	D	A	D	B	A
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	D	C	D	A	D	C	B	A	B	D
Que.	61	62	63	64	65	66				
Ans.	D	B	B	A	C	C				

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	C	A	C	C	C	D	D	D	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	A	B	C	A	D	B	B	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	A	B	C	D	C	C	B	B	C
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	A	B	C	D	C	B	D	A	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	D	A	A	B	C	B	B	A	B	A
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	B	C	A,B,C	A,B,C	B,D	A,C,D	A,C	A,D	A	A,B
Que.	61	62	63	64	65	66	A	B	C	D
Ans.	C,D	D	C	B	D		Q,S	P,S	Q,S	P,S
Que.	67	A	B	C	D	E	68	A	B	C
Ans.		S	S	Q	R	P		R	Q	P

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EXERCISE # S

1. $\frac{5\mu_0 I\theta}{24\pi r}\otimes$ **2.** $5\pi \times 10^{-4}$ T = 1.6×10^{-3} T towards west.

3. (a) zero (b) 2μT along the Z-axis (c) zero and (d) 2μT along the negative Z-axis

4. (a) 0; (b) 1.41×10^{-6} T, 45° in xz-plane, (c) 5×10^{-6} T, +x-direction

5. μ_0 weber.m^{-1}

6. $\frac{\mu_0 nI}{4}(\sqrt{3}+1)$ **7.** 5 **8.** $\frac{2mv_0}{qB}$ **9.** 1.2×10^{-2}m, 4.37×10^{-2}m

10. (a) A horizontal magnetic field of magnitude 0.26 T normal to the conductor in such a direction that Fleming a left-hand rule gives a magnetic force upward.
(b) 1.176 N.

11. (a) Zero,
(b) zero,
(c) force on each electron is evB = IB/(nA) = 5×10^{-25}N. Note : Answer (c) denotes only the magnetic force.

12. (a) 3.1 Nm,
(b) No, the answer is unchanged because the formula $\tau = N\,I\,A \times B$ is true for a planar loop of any shape.

13. $T_0 = 2\pi\sqrt{\frac{m}{6IB}} = 0.57$ s **14.** (a) 1.4, (b) 1

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	1	4	1	2	1	1	3	1	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	2	1	3	1	4	4	3	4	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	2	1	3	2	1	2	4	20	2
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	4	1	4	4	1	1	1	543	1	1
Que.	41	42	43	44	45	46	47			
Ans.	A	C	C	A	B	A	A			

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C	A,D	3	C	B	A,B,C	A	C	D	B
Que.	11	12	13	14	15	16	17			
Ans.	A,B	A,B,D	2 [1.99, 2.01]	4	A,B	4	C			

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